SOLOBOLO / RIEMANN-HYPOTHESIS-FINAL

Hilbert-Vector Resolvents and the Riemann Hypothesis

A detailed analytic seed criterion

Project research note | 28 October 2026

Version 1.3 | CERTIFICATE branch | construction

Authorship, formal verification and invitation to collaborate

This manuscript was written by AI (ChatGPT) under my direction as an independent researcher. It presents my research programme and is offered for scholarly scrutiny, further development and collaboration.

The accompanying framework exists as a Lean implementation and is publicly available in the `Riemann-Hypothesis-Final` GitHub repository, `CERTIFICATE` branch. Identified results have been formally proved and verified in Lean, according to the historical native compilation and axiom-audit records supplied for this project. These include the semantic cancellation-budget equivalence with RH and the first zero-budget cancellation check, $C_0 = 0$. Section 14 identifies the relevant declarations and their verification status.

The verification claim is limited to those documented results. Other arguments remain on-paper derivations or proof-source candidates; universal zero-budget admissibility, the terminating finite-checker equivalence and an unconditional proof of RH remain open.

I am seeking academic endorsement where warranted by independent review, or collaboration with a team of researchers interested in developing this work into a broader, rigorous mathematical theory. I welcome critical examination of the arguments, assistance with the remaining formal proofs, and discussion of the framework's mathematical significance.

Source and research framework

https://github.com/randogelist/Riemann-Hypothesis-Final/tree/CERTIFICATE

Abstract

We develop an analytic Hilbert-vector criterion for the Riemann hypothesis. For $\Theta(w^2) = \xi(\frac{1}{2}+w)$ and $L = \Theta'/\Theta$, RH is equivalent to one bounded nonnegative self-adjoint operator K and one fixed vector v satisfying $(aL(a))' = \|(I+aK)^{-1}v\|^2$ on a positive interval. Resolvent calculus, exact calibration and zero geometry give the forward implication; the actual zero product gives the converse under RH. The earlier derivations of hyperbolic-zeta identities, canonical Jensen polynomials, Bernstein grids, moment-extension intervals and squared-pencil limits are retained. This revision adds the branch's cancellation-budget formulation. If V_n is the total variation of the actual normalized Bernstein row and $C_n = (n+1)(V_{n+1} - V_n)$, its semantic endpoint is RH $\iff \exists N, k \in \mathbb{N}\ \forall n \geq N :\ C_n \leq kV_n$. Historical native compiler evidence supplied by the user reports this equivalence and the first zero-budget check $C_0 = 0$. The pair (N, k) is finite data, but admissibility remains an all-order mathematical predicate, not a terminating Boolean check. Proof sources for the first five cancellation indices remain under native repair; no passing first-five audit is asserted. Universal propagation, finite-checker soundness and completeness, and an accepted finite witness remain open. No unconditional proof of RH or new native Lean audit is claimed.

Evidence and scope. The analytic derivations are on-paper arguments. Section 14 separately identifies exact Lean declarations and historical user-reported native audit evidence; that evidence does not certify every theorem of this paper. The latest first-five build failed, and its partial repair has not been compiled. Publication to `CERTIFICATE` is not a GitHub-CI proof run. The finite-checker target and unconditional RH remain open.

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1 The exact assertion and its role

The starting point is the family of identities

$$R_a = (I + aK)^{-1}, \qquad \frac{\mathrm{d}}{\mathrm{d}a}(aL(a)) = \|R_a v\|^2, \qquad V_a = 2\sqrt{a}\, R_a v. \tag{1.1}$$

Their significance depends on their quantifiers. The operator and vector must be the *same* for every parameter in the interval, and the scalar L must come from the actual completed Riemann zeta function. An arbitrary nonnegative scalar function can be represented as a squared norm by allowing its vector to vary freely. Such a pointwise representation contains none of the resolvent compatibility used below.

We fix the standard completion

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \tag{1.2}$$

with the removable values understood. The symmetry $\xi(s) = \xi(1-s)$ and normalization are classical [1]. Section 2 defines Θ without choosing a square-root branch and proves that L is a well-defined real analytic function on $[0, \infty)$.

Definition 1.1 (Calibrated Hilbert seed)**.** The assertion HSeed_Θ means that there exist a complex Hilbert space $\mathcal{H}$, an operator $K \in \mathcal{B}(\mathcal{H})$, a vector $v \in \mathcal{H}$, and real numbers $0 < \alpha < \beta < \infty$ such that

$$K = K^*, \qquad \langle x, Kx\rangle \geq 0 \quad (x \in \mathcal{H}), \tag{1.3}$$

and, for every $a \in (\alpha, \beta)$,

$$L(a) = \langle v, (I + aK)^{-1} v\rangle. \tag{1.4}$$

The inner product is conjugate-linear in its first slot and linear in its second slot. In particular, the complex extension $z \mapsto \langle v, (I + zK)^{-1}v\rangle$ is holomorphic where the inverse exists.

Definition 1.2 (Differential Hilbert seed)**.** The assertion DSeed_Θ has the same existential data and operator hypotheses as Definition 1.1, but replaces (1.4) by

$$\frac{\mathrm{d}}{\mathrm{d}a}(aL(a)) = \|(I + aK)^{-1}v\|^2 \quad \text{for every } a \in (\alpha, \beta). \tag{1.5}$$

This is an exact functional identity on an interval, not a test at finitely many sample points.

Theorem 1.3 (Main analytic criterion)**.** *For the actual Θ associated with (1.2),*

$$\mathrm{RH} \iff \mathsf{HSeed}_\Theta \iff \mathsf{DSeed}_\Theta. \tag{1.6}$$

If either seed identity holds on one nonempty positive interval, its calibration extends to every $a \geq 0$ for the same K and v. Under RH, a seed can be chosen on $\ell^2(\mathbb{N}; \mathbb{C})$ with K positive, compact and trace class. In that realization,

$$\operatorname{Tr} K = \|v\|^2 = \frac{\xi''(1/2)}{2\xi(1/2)}. \tag{1.7}$$

Sections 3–6 prove the forward and differential implications without assuming RH or a positive zero product. Section 7 proves the converse under RH. The theorem is a criterion: the existential seed assertion carries the mathematical difficulty of RH. No independently constructed seed is supplied here.

There are two different questions behind this note. The analytic question asks exactly what a fixed positive resolvent identity implies. The finite question asks whether a terminating checker on finite raw data can certify such an identity, and whether RH guarantees an accepted input to that same checker. The second question does not follow merely by naming the first one a seed. Section 15 makes this distinction precise.

2 The actual entire function and its zero geometry

2.1 Square descent without a branch choice

Put $E(w) = \xi(\frac{1}{2} + w)$. This is entire, and the functional equation makes it even. Its Taylor series has the form

$$E(w) = \sum_{n=0}^{\infty} c_n w^{2n}, \qquad c_n = \frac{\xi^{(2n)}(1/2)}{(2n)!}. \tag{2.1}$$

Define

$$\Theta(z) = \sum_{n=0}^{\infty} c_n z^n. \tag{2.2}$$

The series has infinite radius of convergence. Indeed, for any $r > 0$, the absolute convergence of (2.1) at a point with $|w| = \sqrt{r}$ gives $\sum_n |c_n| r^n < \infty$. Consequently

$$\Theta(w^2) = \xi(\tfrac{1}{2} + w) \qquad (w \in \mathbb{C}). \tag{2.3}$$

The shorthand $\Theta(z) = \xi(\frac{1}{2} + \sqrt{z})$ therefore has a single-valued meaning: the two square roots give the same value. The branch cut introduced later belongs to the resolvent domain, not to Θ.

Complex conjugation preserves ξ, as follows first on its defining real domain and then by analytic continuation. Thus the coefficients c_n are real, and $\Theta(\overline{z}) = \overline{\Theta(z)}$.

2.2 Positivity on the real positive axis

For completeness, the nonvanishing needed to define L can be checked without RH. For $0 < s < 1$, the alternating-series formula [2, Eq. 25.2.3] gives

$$\zeta(s) = \frac{\eta(s)}{1 - 2^{1-s}}, \qquad \eta(s) = \sum_{n=1}^{\infty} (-1)^{n-1} n^{-s} > 0. \tag{2.4}$$

The strict inequality follows by grouping consecutive terms: $(2j-1)^{-s} - (2j)^{-s} > 0$. The even partial sums converge to the same alternating-series limit and already their first pair is positive. The denominator in (2.4) is negative, so $\zeta(s) < 0$. Since $s(s-1) < 0$ and the remaining factors in (1.2) are positive, $\xi(s) > 0$ on $(0, 1)$. For $s > 1$, every factor in (1.2) is positive, and the removable value at $s = 1$ is $1/2$. Applying this with $s = 1/2 + \sqrt{a}$ proves

$$\Theta(a) > 0 \quad (a \geq 0), \qquad \Theta(0) = \xi(1/2) > 0. \tag{2.5}$$

It follows that

$$L(a) = \frac{\Theta'(a)}{\Theta(a)} \in \mathbb{R} \qquad (a \geq 0) \tag{2.6}$$

is real analytic, including at zero. On this axis it equals the recovered quantity $\mathrm{Re}(\Theta'(a)/\Theta(a))$; taking the real part loses no information there. It would not be legitimate to make that replacement for arbitrary complex z.

Expanding at zero gives the exact normalization

$$\kappa := L(0) = \frac{\Theta'(0)}{\Theta(0)} = \frac{\xi''(1/2)}{2\xi(1/2)}. \tag{2.7}$$

No numerical approximation to κ is needed in any proof below.

2.3 The critical line becomes the negative real axis

The zeros of ξ are exactly the nontrivial zeros of ζ; the standard zero symmetry and the meaning of RH are recalled in [3]. If $\rho = 1/2 + w$ is such a zero, then $z = w^2$ is a zero of Θ, and conversely every zero of Θ arises this way by (2.3).

For $w = x + iy$,

$$w^2 = (x^2 - y^2) + 2ixy. \tag{2.8}$$

If w^2 is real and nonpositive, then $xy = 0$ and $x^2 - y^2 \le 0$. If $y = 0$, these imply $x = 0$; if $y \neq 0$, the equality $xy = 0$ again implies $x = 0$. Conversely $x = 0$ gives $w^2 = -y^2 \le 0$. Since $\Theta(0) \neq 0$, its zeros never correspond to $w = 0$. We have therefore proved

Proposition 2.1 (Exact zero geometry)**.**

$$\mathrm{RH} \quad \Longleftrightarrow \quad \big(\Theta(z) = 0 \implies z \in (-\infty, 0)\big). \tag{2.9}$$

This elementary change of variables is the final geometric step of the forward proof. It refers to the actual ξ and Θ throughout.

3 Resolvent calculus with one fixed vector

Fix a complex Hilbert space $\mathcal{H}$, a bounded nonnegative self-adjoint operator K, and a vector v. For $a \ge 0$, set

$$R_a = (I + aK)^{-1}, \qquad f(a) = \langle v, R_a v\rangle, \qquad g(a) = af(a). \tag{3.1}$$

At this stage f is a model scalar function; equality with the actual L has not been assumed.

3.1 Invertibility, self-adjointness and commutation

For every $x \in \mathcal{H}$, positivity and Cauchy–Schwarz imply

$$\|x\|^2 \le \langle x, (I + aK)x\rangle \le \|x\|\,\|(I + aK)x\|.$$

Thus $I + aK$ is bounded below by 1. Its range is closed. Its orthogonal complement is the kernel of its adjoint, which equals its own kernel because the operator is self-adjoint. The lower bound makes that kernel zero, so the range is also dense and hence is all of $\mathcal{H}$. It follows that

$$\|R_a\| \le 1, \qquad R_a^* = R_a, \qquad R_a \ge 0. \tag{3.2}$$

For the last inequality write $y = (I + aK)x$ and observe that $\langle y, R_a y\rangle = \langle (I + aK)x, x\rangle \ge 0$. The operator K commutes with $I + aK$ and therefore with its inverse. The operators R_a and R_b also commute.

Lemma 3.1 (Two resolvent identities)**.** *For $a, b \ge 0$,*

$$R_a - R_b = (b - a)R_a K R_b, \tag{3.3}$$

$$aR_a - bR_b = (a - b)R_a R_b. \tag{3.4}$$

Proof. For the first identity, multiply out $R_a[(I + bK) - (I + aK)]R_b$. For the second, the same calculation with $a(I + bK) - b(I + aK) = (a - b)I$ gives the result. Both formulas also hold when $a = b$, with both sides then zero. □

3.2 Differentiation in operator norm

From (3.3),

$$\frac{R_{a+h} - R_a}{h} = -R_{a+h} K R_a.$$

Continuity of inversion, or the same identity and local boundedness, shows that the right side tends in operator norm to $-R_a K R_a$. Since K commutes with R_a,

$$R_a' = -K R_a^2. \tag{3.5}$$

Repeated differentiation yields

$$R_a^{(n)} = (-1)^n n! K^n R_a^{n+1} \qquad (n \geq 0). \tag{3.6}$$

One can verify the induction directly: differentiating R_a^{n+1} produces $n+1$ identical commuting terms $-K R_a^{n+2}$. Applying the continuous linear functional $T \mapsto \langle v, Tv \rangle$ gives

$$(-1)^n f^{(n)}(a) = n! \langle v, K^n R_a^{n+1} v \rangle \geq 0. \tag{3.7}$$

The last operator is positive, for example by the bounded self-adjoint functional calculus [6]. Thus f is completely monotone. This consequence alone will not characterize a calibrated seed.

3.3 The norm-square identity

The identity $(I + aK)R_a = I$ implies $R_a = R_a^2 + aK R_a^2$. Consequently

$$\begin{aligned} g'(a) &= f(a) + a f'(a) \\ &= \langle v, (R_a - aK R_a^2) v \rangle \\ &= \langle v, R_a^2 v \rangle = \|R_a v\|^2. \end{aligned} \tag{3.8}$$

This proves the central algebraic formula, including its exact coefficient. It involves $af(a)$, not $f(a)$ alone and not an unspecified change of basepoint. Differentiating further gives

$$(-1)^n g^{(n+1)}(a) = (n+1)! \langle v, K^n R_a^{n+2} v \rangle \geq 0 \qquad (n \geq 0). \tag{3.9}$$

In particular g is increasing and concave on the positive axis. If $v \neq 0$, then $R_a v \neq 0$ and $g'(a) > 0$.

4 The Gram kernel and its exact normalization

Define $V_a = 2\sqrt{a}\, R_a v$ for $a > 0$. Taking the scalar product of (3.4) with v and using self-adjointness gives, for $a \neq b$,

$$\frac{g(a) - g(b)}{a - b} = \langle v, R_a R_b v \rangle = \langle R_a v, R_b v \rangle. \tag{4.1}$$

Hence

$$\langle V_a, V_b \rangle = 4\sqrt{ab}\, \frac{a f(a) - b f(b)}{a - b} \qquad (a, b > 0,\ a \neq b). \tag{4.2}$$

If the calibration $f = L$ holds, this is precisely the recovered Hilbert-vector Gram identity.

Proposition 4.1 (Diagonal and positivity)**.** *The divided-difference kernel*

$$\mathcal{K}_g(a,b) = \begin{cases} \dfrac{g(a)-g(b)}{a-b}, & a \neq b, \\ g'(a), & a = b \end{cases} \tag{4.3}$$

is positive semidefinite. More precisely, for any finite collection $a_1, \dots, a_m > 0$ and complex numbers $c_1, \dots, c_m$,

$$\sum_{i,j=1}^{m} \overline{c_i} c_j \mathcal{K}_g(a_i, a_j) = \|\sum_{j=1}^{m} c_j R_{a_j} v\|^2 \geq 0. \tag{4.4}$$

The diagonal extension of (4.2) is

$$\|V_a\|^2 = 4a\, g'(a) = 4a\, \|R_a v\|^2. \tag{4.5}$$

Proof. Equation (4.1) gives all off-diagonal entries, while (3.8) gives the diagonal. Expanding the displayed squared norm proves (4.4); multiplication by the factors $2\sqrt{a_i}$ gives the scaled Gram matrix. □

For two distinct parameters, the determinant condition is the concrete inequality

$$\left|\frac{g(a)-g(b)}{a-b}\right|^2 \leq g'(a)g'(b). \tag{4.6}$$

It is a necessary condition tying different points together. Positivity of g' at every individual point does not imply it.

Example 4.2 (Even strong scalar positivity is insufficient)**.** Let $\widetilde{g}(a) = 1 - e^{-a}$ and $\widetilde{f}(a) = (1-e^{-a})/a$ for $a > 0$. Then $\widetilde{g}' > 0$, $\widetilde{g}'' < 0$, and

$$\widetilde{f}(a) = \int_0^1 e^{-at}\,\mathrm{d}t, \qquad (-1)^n \widetilde{f}^{(n)}(a) = \int_0^1 t^n e^{-at}\,\mathrm{d}t > 0.$$

Nevertheless, writing $d = a - b \neq 0$, the off-diagonal divided difference is

$$\mathcal{K}_{\widetilde{g}}(a,b) = e^{-(a+b)/2}\frac{\sinh(d/2)}{d/2}.$$

The determinant of the corresponding two-by-two matrix equals

$$e^{-a-b}\left[1 - \left(\frac{\sinh(d/2)}{d/2}\right)^2\right] < 0. \tag{4.7}$$

The strict inequality follows from $|\sinh x| > |x|$ for nonzero real x, which follows by integrating $\cosh t > 1$ away from zero. Thus these scalar monotonicity conditions cannot be substituted for the fixed-resolvent structure.

5 Holomorphic extension of the model

Choose a real number $M > \|K\|$ with $M > 0$, and set

$$D_M = \mathbb{C} \setminus (-\infty, -1/M]. \tag{5.1}$$

This open connected domain contains zero and the entire positive real axis. The bounded self-adjoint spectral theorem puts $\sigma(K)$ in $[0, M]$ [6]. For $z \in D_M$ and $t \in [0, M]$, the number $1 + zt$ is nonzero. Indeed, a zero would require $t > 0$ and $z = -1/t \leq -1/M$. The functional calculus therefore defines

$$R_z = (I + zK)^{-1}, \qquad F(z) = \langle v, R_z v\rangle \qquad (z \in D_M). \tag{5.2}$$

Lemma 5.1 (Norm-holomorphic resolvent)**.** *The maps $z \mapsto R_z$ and $z \mapsto F(z)$ are holomorphic on D_M.*

Proof. Fix $z_0 \in D_M$. For sufficiently small h, $\|hKR_{z_0}\| < 1$ and

$$I + (z_0 + h)K = (I + z_0K)(I + hKR_{z_0}),$$
$$R_{z_0+h} = R_{z_0} \sum_{n=0}^{\infty} (-hKR_{z_0})^n.$$

The series converges in operator norm on a neighborhood of $h = 0$, so it is a norm-convergent local power series. The scalar functional $T \mapsto \langle v, Tv\rangle$ is continuous and complex linear under the convention in Definition 1.1, and applying it proves the scalar assertion. □

It is useful to retain zero in D_M. A cut at $(-\infty, 0]$ would still suffice for the normalized seed-to-RH argument, but it would hide the evaluation at zero that removes the integration constant in the differential seed. Boundedness of K supplies a genuine neighborhood of zero on which the resolvent is analytic.

The spectral theorem also gives a scalar measure. If E_K is the projection-valued spectral measure, define

$$\mu(B) = \langle v, E_K(B)v\rangle \qquad (B \text{ Borel in } [0, M]). \tag{5.3}$$

It is a finite positive measure, with total mass $\|v\|^2$, and

$$F(z) = \int_{[0,M]} \frac{1}{1 + zt}\, \mathrm{d}\mu(t). \tag{5.4}$$

For a compact $C \subset D_M$, the continuous function $|1 + zt|$ has a strictly positive minimum on $C \times [0, M]$. This bounds every fixed derivative of the integrand by an integrable constant. Thus

$$F^{(n)}(z) = (-1)^n n! \int_{[0,M]} \frac{t^n}{(1 + zt)^{n+1}}\, \mathrm{d}\mu(t). \tag{5.5}$$

This justifies the integral differentiations directly and independently checks the operator formulas of Section 3.

6 The analytic transfer to RH

6.1 Exact calibration on an interval is enough

Theorem 6.1 (Normalized seed implies RH)**.** *If HSeed_Θ holds, then* RH *holds. For its fixed seed data, $F(a) = L(a)$ for all $a \geq 0$.*

Proof. On the interval in Definition 1.1, the calibration and $\Theta(a) > 0$ imply

$$\Theta'(a) - F(a)\Theta(a) = 0.$$

The function

$$G(z) = \Theta'(z) - F(z)\Theta(z) \tag{6.1}$$

is holomorphic on D_M. Its zeros have an accumulation point there, so the identity theorem gives

$$\Theta'(z) = F(z)\Theta(z) \qquad (z \in D_M). \tag{6.2}$$

This passage does not assume that Θ is zero-free on D_M. The expression (6.1) has no division by Θ.

Suppose, for a contradiction, that Θ has a zero $z_0 \in D_M$. Since $\Theta(0) > 0$, it is not identically zero. Its zero at z_0 has some finite order $m \geq 1$, so near z_0,

$$\Theta(z) = (z - z_0)^m h(z), \qquad h(z_0) \neq 0,$$

where h is holomorphic. Substituting in (6.2) and cancelling $(z - z_0)^{m-1}$ off z_0 gives

$$mh(z) + (z - z_0)h'(z) = (z - z_0)F(z)h(z).$$

Both sides are continuous at z_0; evaluation there yields $mh(z_0) = 0$, a contradiction. Therefore every zero of Θ lies outside D_M, hence in $(-\infty, -1/M] \subset (-\infty, 0)$. Proposition 2.1 gives RH. Finally, restricting (6.2) to $a \geq 0$ and dividing by the already known positive $\Theta(a)$ proves $F(a) = L(a)$ everywhere on that ray. □

The proof uses neither a preselected logarithm of Θ on the cut plane nor an RH-dependent product. Choosing a global logarithm before proving zero-freeness would conceal the central difficulty. The division-free ODE avoids that issue.

6.2 The integration constant is real and must be treated

Suppose only the differential identity (1.5) holds. From (3.8),

$$\frac{\mathrm{d}}{\mathrm{d}a}(aL(a) - af(a)) = 0 \quad (a \in (\alpha, \beta)).$$

The ordinary mean value theorem on subintervals shows that

$$a(L(a) - F(a)) = C, \qquad L(a) = F(a) + \frac{C}{a} \tag{6.3}$$

for a constant $C \in \mathbb{R}$. Neither the derivative identity nor the off-diagonal Gram identity alone detects this constant: adding C to $aL(a)$ changes neither its derivative nor its divided differences.

For an arbitrary function on an interval bounded away from zero, one would need a normalization value to set $C = 0$. The actual entire Θ supplies a stronger analytic normalization.

Theorem 6.2 (Differential seed and normalized seed agree)**.** *For the actual Θ, $\mathsf{DSeed}_\Theta \iff \mathsf{HSeed}_\Theta$. The same operator and vector can be used in both directions.*

Proof. If HSeed_Θ holds, differentiating $aL(a) = af(a)$ and using (3.8) gives DSeed_Θ. Conversely assume DSeed_Θ and let C be the constant in (6.3). Consider

$$Q(z) = z\Theta'(z) - (zF(z) + C)\Theta(z). \tag{6.4}$$

This is holomorphic on all of D_M, including zero. By (6.3), it vanishes on (α, β), so the identity theorem gives $Q = 0$ throughout D_M. At zero,

$$0 = Q(0) = -C\Theta(0).$$

Equation (2.5) therefore forces $C = 0$. It follows that $z(\Theta' - F\Theta) = 0$ on D_M. Away from zero we may divide by z, and continuity supplies the equality at zero. Thus (6.2) holds, and positive-ray division by $\Theta(a)$ gives the normalized calibration. In particular HSeed_Θ holds. □

Corollary 6.3 (The recovered norm-square identity is an RH criterion)**.** *For a fixed bounded nonnegative self-adjoint K and fixed v, the identity (1.5) on any nonempty positive interval implies RH. No extra origin-normalization premise is needed beyond the established facts about the actual Θ.*

The qualifier "actual" is essential. If one replaced Θ by $z^m\Theta$ with $m > 0$, its logarithmic derivative would acquire m/z, and the derivative of aL would not see that origin factor. The known value $\Theta(0) \neq 0$ is precisely what rules out this ambiguity here.

7 The converse: constructing a seed under RH

7.1 Why the product has no exponential ambiguity

We use two classical facts about entire functions. The completed ξ has order at most 1; a direct growth argument from the theta integral is given in [4]. Hadamard factorization and its zero-summability consequence are stated in [5, Theorems 3.7 and 3.11]. To see exactly how these apply, write $M_E(r) = \max_{|w|=r} |E(w)|$ and define M_Θ similarly. The map $w \mapsto w^2$ maps the circle of radius $\sqrt{r}$ onto the circle of radius r, so

$$M_\Theta(r) = M_E(\sqrt{r}). \tag{7.1}$$

A bound $|E(w)| \le C_\varepsilon \exp(C_\varepsilon |w|^{1+\varepsilon})$ therefore puts the order of Θ at most $1/2$. Only the strict inequality "order less than 1" is needed below.

For an entire function of order less than 1, nonzero at zero, Hadamard factorization has genus zero and a constant exponential factor. Its nonzero zeros z_n, repeated by multiplicity, satisfy $\sum_n |z_n|^{-1} < \infty$. Hence

$$\Theta(z) = \Theta(0) \prod_n \left(1 - \frac{z}{z_n}\right). \tag{7.2}$$

There is no factor e^{bz} with $b \neq 0$: the polynomial in the Hadamard exponential has degree at most the order, hence degree zero. There is also no origin monomial because $\Theta(0) \neq 0$.

7.2 Zero pairs and multiplicity

Assume RH. Enumerate the positive ordinates of the nontrivial zeros as $\gamma_1, \gamma_2, \ldots$, repeating each according to its multiplicity. There are infinitely many zeros, with no finite accumulation point [3]. We can arrange $0 < \gamma_1 \le \gamma_2 \le \cdots$ with $\gamma_n \to \infty$. The two points $w = \pm i\gamma$ are the preimages of $z = -\gamma^2$. Since the derivative of $w \mapsto w^2$ is nonzero at either preimage, a zero of multiplicity m at either point corresponds to a zero of multiplicity m of Θ. It does *not* become a zero of multiplicity $2m$. Equation (7.2) becomes

$$\Theta(z) = \Theta(0) \prod_{n=1}^{\infty} \left(1 + \frac{z}{\gamma_n^2}\right), \qquad \sum_{n=1}^{\infty} \frac{1}{\gamma_n^2} < \infty. \tag{7.3}$$

Thus one positive/negative pair contributes one factor for each occurrence of its multiplicity.

7.3 Logarithmic differentiation with convergence controlled

Fix a compact set disjoint from the negative zero set, and enclose it in $|z| \le r$. For all sufficiently large n, $\gamma_n^2 > 2r$ and

$$\left|\frac{1}{z + \gamma_n^2}\right| \le \frac{2}{\gamma_n^2}.$$

The right side is summable by (7.3). Finite initial factors have no zeros on the chosen compact set. Locally uniform convergence of the product and its logarithmic derivatives therefore justifies

$$\frac{\Theta'(z)}{\Theta(z)} = \sum_{n=1}^{\infty} \frac{1}{z + \gamma_n^2} \tag{7.4}$$

away from the zeros. In particular this identity holds at every $a \ge 0$.

Proposition 7.1 (Diagonal realization under RH)**.** *Let $\mathcal{H} = \ell^2(\mathbb{N}; \mathbb{C})$, and define*

$$(Kx)_n = \gamma_n^{-2} x_n, \qquad v_n = \gamma_n^{-1}. \tag{7.5}$$

Then K is bounded, self-adjoint, nonnegative, compact and trace class; $v \in \mathcal{H}$; and

$$\langle v, (I + aK)^{-1} v \rangle = L(a) \qquad (a \geq 0). \tag{7.6}$$

Proof. The positive ordinates have a positive minimum: a sequence tending to zero would give an accumulation of zeros at $1/2$, contradicting $\xi(1/2) > 0$. Hence the diagonal entries of K are bounded. They are positive real numbers and tend to zero, which gives self-adjointness, positivity and compactness. Their sum is finite by (7.3), so the positive diagonal operator is trace class. The same sum equals $\sum_n |v_n|^2$, proving $v \in \mathcal{H}$. Finally,

$$\langle v, (I + aK)^{-1} v \rangle = \sum_n \frac{\gamma_n^{-2}}{1 + a\gamma_n^{-2}} = \sum_n \frac{1}{a + \gamma_n^2} = L(a),$$

using (7.4). □

This proves RH $\Rightarrow \mathsf{HSeed}_\Theta$, completing Theorem 1.3. Taking $a = 0$ gives the normalization

$$\operatorname{Tr} K = \|v\|^2 = \sum_n \gamma_n^{-2} = L(0) = \frac{\xi''(1/2)}{2\xi(1/2)}. \tag{7.7}$$

For an arbitrary calibrated representation, $\|v\|^2 = L(0)$ still holds. The equality with $\operatorname{Tr} K$ is asserted only for the particular diagonal construction just given; arbitrary invisible operator components need not have that trace.

If one prefers distinct positive ordinates γ_j with multiplicities m_j, the equivalent scalar model uses $K_{jj} = \gamma_j^{-2}$ and $v_j = \sqrt{m_j}/\gamma_j$. Neither version assumes that zeta zeros are simple. In the repeated model, the vector need not be cyclic when an eigenvalue is repeated. No cyclicity assumption is needed for the argument.

8 Spectral measures, moments and exact reconstruction

8.1 A scalar measure formulation

Define MSeed_Θ to mean that for some $M > 0$ and finite positive Borel measure μ on $[0, M]$,

$$L(a) = \int_{[0,M]} \frac{1}{1 + at} \, \mathrm{d}\mu(t) \tag{8.1}$$

on a nonempty positive interval. Section 5 proves $\mathsf{HSeed}_\Theta \Rightarrow \mathsf{MSeed}_\Theta$. Conversely, given (8.1), take

$$\mathcal{H} = L^2([0, M], \mu), \qquad (Ku)(t) = tu(t), \qquad v(t) = 1.$$

Finite mass makes v square integrable, while multiplication by t is bounded, nonnegative and self-adjoint. Its scalar resolvent is exactly (8.1). Therefore

$$\mathsf{MSeed}_\Theta \iff \mathsf{HSeed}_\Theta \iff \text{RH}. \tag{8.2}$$

The word "finite" in *finite measure* means finite total mass. It does not mean finite support, a finite list of rational numbers, or a finite-checker witness. Under RH the measure is

$$\mu = \sum_n \gamma_n^{-2} \delta_{\gamma_n^{-2}}, \qquad \mu([0, M]) = \kappa. \tag{8.3}$$

This differs from the zero-counting measure $\sum_n \delta_{\gamma_n^2}$, which has infinite total mass. The reciprocal variable and the weights are essential to the normalization.

8.2 Moments and positive finite sections

For any fixed positive operator model, put

$$m_n = \langle v, K^n v\rangle = \int_{[0,M]} t^n \, \mathrm{d}\mu(t) \qquad (n \geq 0). \tag{8.4}$$

If $|z|M < 1$, the Neumann series converges uniformly in the spectral variable and gives

$$F(z) = \sum_{n=0}^{\infty} (-1)^n m_n z^n, \qquad 0 \leq m_n \leq M^n m_0. \tag{8.5}$$

The remainder after degree N has the exact form

$$F(z) - \sum_{n=0}^{N} (-1)^n m_n z^n = (-z)^{N+1} \int \frac{t^{N+1}}{1+zt} \, \mathrm{d}\mu(t), \tag{8.6}$$

and consequently

$$\left| F(z) - \sum_{n=0}^{N} (-1)^n m_n z^n \right| \leq \frac{m_0 (M|z|)^{N+1}}{1 - M|z|}. \tag{8.7}$$

For a calibrated model,

$$m_n = \frac{(-1)^n}{n!} L^{(n)}(0). \tag{8.8}$$

This is an all-orders identity for actual derivatives.

For a polynomial $p(t) = \sum_i c_i t^i$, positivity of μ gives

$$\sum_{i,j} \overline{c_i} c_j m_{i+j} \geq 0, \tag{8.9}$$

$$\sum_{i,j} \overline{c_i} c_j m_{i+j+1} \geq 0, \tag{8.10}$$

$$\sum_{i,j} \overline{c_i} c_j (M^2 m_{i+j} - m_{i+j+2}) \geq 0. \tag{8.11}$$

These are the integrals of $|p|^2$, $t|p|^2$ and $(M^2 - t^2)|p|^2$. Every finite matrix in this hierarchy is positive semidefinite. The following converse records exactly what an *all-orders* moment construction can accomplish.

Theorem 8.1 (Bounded positive moment reconstruction)**.** *Let $(m_n)_{n\geq 0}$ be a real sequence and $M > 0$. Suppose (8.9)–(8.11) hold for every finite coefficient list. Then there exist a Hilbert space, a bounded nonnegative self-adjoint operator K with $\|K\| \leq M$, and a vector v such that $m_n = \langle v, K^n v\rangle$ for all n.*

Proof. Define a complex linear functional ℓ on polynomials by $\ell(t^n) = m_n$, and a sesquilinear form by $\langle p, q\rangle_0 = \ell(\overline{p}\, q)$, where conjugation acts on coefficients and t is real. Condition (8.9) makes this form positive semidefinite, hence it satisfies Cauchy–Schwarz. Quotient the polynomial space by the vectors of zero seminorm and complete to a Hilbert space $\mathcal{H}$.

Multiplication by t is well defined on the quotient: by (8.11),

$$\|tp\|_0^2 = \ell(t^2|p|^2) \leq M^2 \ell(|p|^2) = M^2 \|p\|_0^2.$$

Thus it sends every null vector to a null vector and is bounded by M. It extends to an everywhere defined bounded operator K on $\mathcal{H}$. Since $\ell(\overline{tp}\, q) = \ell(\overline{p}\, tq)$, it is symmetric on a dense subspace and hence self-adjoint after extension. Condition (8.10) gives $\langle [p], K[p]\rangle = \ell(t|p|^2) \geq 0$; continuity makes K positive on all of $\mathcal{H}$. Let $v = [1]$. Then $K^n v = [t^n]$ and $\langle v, K^n v\rangle = \ell(t^n) = m_n$, as required. □

Corollary 8.2 (An all-order actual-jet criterion)**.** *Define m_n by (8.8). RH is equivalent to the existence of $M > 0$ such that (8.9)–(8.11) hold for every finite coefficient list.*

Proof. Under RH, the diagonal seed supplies these inequalities. Conversely, Theorem 8.1 constructs K, v with the specified moments. The resolvent has Taylor series (8.5). The actual L is analytic near zero and has exactly the same Taylor coefficients. They therefore agree on a common disk of positive radius, and in particular on a positive open interval. This is HSeed_Θ, which implies RH. □

The quantifier "for every finite coefficient list" ranges over unbounded degrees. Checking finitely many matrices does not establish this corollary. A finite certificate would have to derive the entire hierarchy by an independently proved uniform mechanism.

8.3 What is determined by calibration

If two compactly supported finite positive measures represent the same actual L on an interval, analytic continuation gives equality near zero. Their moments consequently agree. View both on a common compact interval containing their supports. Polynomials are dense in the continuous functions on that interval, and equality of integrals for polynomials extends, by the total-mass bound, to all continuous functions. Uniqueness of finite Borel measures then gives equality of the two measures. Thus the scalar spectral measure seen by v is determined. This does not identify every operator on the ambient Hilbert space: an orthogonal summand invisible to v leaves all the scalar identities unchanged.

9 Basepoint shifts and finite truncations

9.1 Two different shifted vectors

Fix $a_0 \geq 0$, and abbreviate $R_0 = (I + a_0K)^{-1}$ in this subsection only. Set

$$B = KR_0, \qquad w = R_0^{1/2}v, \qquad u = R_0v. \tag{9.1}$$

These two vectors have different purposes. For $t = a - a_0$ with $a \geq 0$,

$$I + aK = (I + a_0K)(I + tB).$$

All the displayed operators commute. Therefore

$$f(a_0 + t) = \langle w, (I + tB)^{-1}w\rangle. \tag{9.2}$$

The functional calculus gives $0 \leq B \leq M(1 + a_0M)^{-1}I$ whenever $0 \leq K \leq MI$. In particular the local shifted inverse is analytic at $t = 0$.

On the other hand, (3.4) gives

$$g(a_0 + t) - g(a_0) = t\langle u, (I + tB)^{-1}u\rangle. \tag{9.3}$$

To verify this, write $aR_a - a_0R_0 = tR_aR_0$ and $R_a = (I + tB)^{-1}R_0$, then take the scalar product with v. Differentiating (9.3) gives

$$g'(a_0 + t) = \|(I + tB)^{-1}u\|^2. \tag{9.4}$$

The additive value $g(a_0)$ and the distinction $w \neq u$ in general must be retained. Merely replacing a by $a - a_0$ in part of a formula changes the mathematical condition.

9.2 What finite diagonal truncations do prove

Under RH, use the repeated-ordinate model from Proposition 7.1, and truncate after N entries:

$$F_N(z) = \sum_{n=1}^{N} \frac{1}{z+\gamma_n^2}, \qquad S_N = \sum_{n>N} \gamma_n^{-2}.$$

For every $a \geq 0$,

$$0 \leq L(a) - F_N(a) \leq S_N. \tag{9.5}$$

To state a bound on a whole disk, define the tail directly by

$$T_N(z) = \sum_{n>N} \frac{1}{z+\gamma_n^2}.$$

If $\gamma_{N+1}^2 > 2r$, this series is holomorphic on a neighborhood of $|z| \leq r$. Away from the initial poles it equals $F - F_N$; at a shared pole among the first N terms, it is the holomorphic extension of that difference after cancellation. Thus, on the entire disk,

$$|T_N(z)| \leq 2S_N. \tag{9.6}$$

For each derivative order $j \geq 0$, the same argument gives

$$|T_N^{(j)}(z)| \leq j!\, 2^{j+1} \sum_{n>N} \gamma_n^{-2j-2}. \tag{9.7}$$

These estimates follow term by term from the convergent series and its displayed denominator bounds. They show what a tail certificate would need to control, including its dependence on derivative order.

The construction here assumes RH to identify the positive spectrum. It is consequently not an unconditional finite computation proving RH. Even a uniform small error does not imply exact equality; a uniform error bound for values alone does not automatically supply an error bound for all derivatives on the same domain.

10 Hyperbolic zeta identities and the actual-seed transfer

The hyperbolic zeta identities provide explicit positive measures from which the operator calculus of Section 3 can be constructed. This section makes that connection precise, including the normalization of the odd-zeta formulas. It also identifies a limitation: the resulting model functions have a different large-parameter behavior from the actual L. Their positivity is useful structural information, but these particular models cannot themselves be its exact calibration. We then derive an actual-ξ Mellin identity and a conditional coefficient-limit theorem for positive determinant models obtained from the project's signed pencils. The limit condition is stated as an open analytic obligation.

The source identities below are traced to the manuscript [11] and repository revision `b56c810` [12, 13, 14, 16, 17]. Only the displayed identities are used, with proofs or explicit classical inputs supplied here. Source inspection is separate from native compilation and axiom auditing.

10.1 A positive weight and its Mellin moments

For an integer $k \geq 0$, set

$$B(x) = \frac{1}{e^x - 1}, \qquad \omega_k(y) = \frac{2(2\pi)^{2k}}{(2k)!} y^{2k+1} B^{(2k)}(2\pi y), \qquad y > 0. \tag{10.1}$$

Here $B^{(2k)}$ denotes differentiation with respect to the argument of B, not with respect to y. This distinction determines the factor $(2\pi)^{2k}$. The weights and their moment identities are the relevant positive-measure objects in the hyperbolic zeta development [12, 13]. If a continuous weight parameter $s > 1$ is used, the coefficient extending these weights is $2(2\pi)^{s-1}/\Gamma(s)$. The printed continuous coefficient $(2\pi)^s/(2\Gamma(s))$ differs by $\pi/2$; the repository records this correction explicitly [15].

Proposition 10.1 (Positivity and exact moments)**.** *The function ω_k is strictly positive, tends to $1/\pi$ at zero, and defines a finite measure on $(0,\infty)$. For every real $q > -1$,*

$$\int_0^\infty y^q \omega_k(y)\,\mathrm{d}y = \frac{2\Gamma(q+2k+2)\zeta(q+2)}{(2k)!(2\pi)^{q+2}}. \tag{10.2}$$

In particular,

$$\int_0^\infty \omega_k(y)\,\mathrm{d}y = \frac{2k+1}{12}. \tag{10.3}$$

Proof. The geometric series and its locally uniformly convergent derivatives give

$$B^{(2k)}(x) = \sum_{n=1}^{\infty} n^{2k} e^{-nx} > 0 \qquad (x > 0).$$

The Laurent expansion $B(x) = x^{-1} - 1/2 + O(x)$, with its analytic remainder near zero, shows $B^{(2k)}(x) \sim (2k)!x^{-2k-1}$. Substitution in (10.1) gives $\omega_k(y) \to 1/\pi$. At infinity the differentiated geometric series is $O(e^{-x})$, so the weight has exponential decay times a polynomial.

For the exact value, Tonelli's theorem applies because every summand is nonnegative. The gamma integral then yields

$$\begin{aligned}\int_0^\infty y^q \omega_k(y)\,\mathrm{d}y &= \frac{2(2\pi)^{2k}}{(2k)!}\sum_{n\ge1} n^{2k}\frac{\Gamma(q+2k+2)}{(2\pi n)^{q+2k+2}} \\ &= \frac{2\Gamma(q+2k+2)}{(2k)!(2\pi)^{q+2}}\sum_{n\ge1} n^{-q-2}.\end{aligned}$$

The last series converges precisely in the stated real range. Taking $q = 0$ and using $\zeta(2) = \pi^2/6$ proves (10.3). □

Thus polynomial Gram matrices have exact entries involving even and odd zeta values. Their positivity alone does not make multiplication by y or y^2 bounded: the measure has unbounded support. The next construction supplies the bounded operator required by the seed framework. For instance, the even moments $b_j = \int y^{2j}\omega_k(y)\,\mathrm{d}y$ satisfy

$$\frac{b_{j+1}}{b_j} = \frac{(2k+2j+3)(2k+2j+2)}{4\pi^2}\frac{\zeta(2j+4)}{\zeta(2j+2)} \longrightarrow \infty. \tag{10.4}$$

Thus no fixed finite support bound can be read into the raw moment sequence. Positivity and bounded support are different requirements in the reconstruction theorem. Here the support is made compact by an explicit change of spectral variable, whose effect on the vector and measure must also be included.

10.2 Compactification and the fixed Hilbert vector

Put

$$t(y) = \frac{1}{1+y^2}, \qquad \mathcal{H}_k = L^2((0,\infty), \omega_k(y)\,\mathrm{d}y), \qquad (K_k u)(y) = t(y)u(y), \qquad v_k(y) = \sqrt{t(y)}. \tag{10.5}$$

Then K_k is nonnegative and self-adjoint, with $\|K_k\| = 1$. Indeed, $0 < t(y) < 1$, and values arbitrarily close to 1 occur on sets of positive measure. Also $v_k \in \mathcal{H}_k$ by (10.3). Its scalar resolvent is

$$F_k(a) = \langle v_k, (I + aK_k)^{-1} v_k \rangle = \int_0^\infty \frac{\omega_k(y)}{a + 1 + y^2}\, \mathrm{d}y, \qquad a > -1. \tag{10.6}$$

In particular, $\|v_k\|^2 = F_k(0)$; it is not the total mass in (10.3). The factor $\sqrt{t(y)}$ in the vector is essential to this normalization: its square supplies the numerator $t(y)$ in $t(y)/(1 + at(y)) = (a+1+y^2)^{-1}$. All parameters consequently use one operator and one vector, exactly as required in the seed definitions. The construction is infinite dimensional, despite the single integer parameter specifying the density.

Define the finite measure ν_k on $[0, 1]$ by

$$\int \varphi(t)\, \mathrm{d}\nu_k(t) = \int_0^\infty t(y)\varphi(t(y))\omega_k(y)\, \mathrm{d}y.$$

Its moments are

$$q_n^{(k)} = \int_0^1 t^n\, \mathrm{d}\nu_k(t) = \int_0^\infty \frac{\omega_k(y)}{(1 + y^2)^{n+1}}\, \mathrm{d}y, \qquad F_k(a) = \sum_{n \geq 0} (-1)^n q_n^{(k)} a^n \quad (|a| < 1). \tag{10.7}$$

The series follows from a uniformly convergent geometric series on $0 \leq t \leq 1$. For every polynomial $p(t) = \sum_i c_i t^i$, integration of $|p|^2$, $t|p|^2$, $(1 - t)|p|^2$ and $(1 - t^2)|p|^2$ proves positivity of all finite sections of

$$(q_{i+j}^{(k)}), \quad (q_{i+j+1}^{(k)}), \quad (q_{i+j}^{(k)} - q_{i+j+1}^{(k)}), \quad (q_{i+j}^{(k)} - q_{i+j+2}^{(k)}). \tag{10.8}$$

These are all-order identities for this explicitly defined model. They make no identification with the actual moments $(-1)^n L^{(n)}(0)/n!$. There is also the full compact-moment difference identity

$$\sum_{j=0}^{r} (-1)^j \binom{r}{j} q_{n+j}^{(k)} = \int_0^1 t^n (1 - t)^r\, \mathrm{d}\nu_k(t) \geq 0 \qquad (n, r \geq 0). \tag{10.9}$$

For a nonzero polynomial, each quadratic form in (10.8) is strictly positive: its integrand is positive except at finitely many roots and possibly the endpoints, and the transformed measure gives positive mass to every open subinterval of $(0, 1)$.

The compact moments should not be obtained by formally expanding $(1+y^2)^{-1}$ into powers of y^2 and integrating term by term. That geometric series does not converge throughout the original unbounded support. Formula (10.7), or the Hurwitz calculation below, supplies a justified route instead.

With $R_{k,a} = (I + aK_k)^{-1}$ and $g_k(a) = aF_k(a)$, the general resolvent calculation gives the explicit formulas

$$g_k'(a) = \|R_{k,a} v_k\|^2 = \int_0^\infty \frac{(1 + y^2)\omega_k(y)}{(a + 1 + y^2)^2}\, \mathrm{d}y, \tag{10.10}$$

$$\frac{g_k(a) - g_k(b)}{a - b} = \langle R_{k,a} v_k, R_{k,b} v_k \rangle = \int_0^\infty \frac{(1 + y^2)\omega_k(y)}{(a + 1 + y^2)(b + 1 + y^2)}\, \mathrm{d}y. \tag{10.11}$$

The second formula is for distinct nonnegative a, b and extends diagonally by the first. Multiplication by $4\sqrt{ab}$ produces the Gram identity for $2\sqrt{a}\, R_{k,a} v_k$.

10.3 The Hurwitz identity and its endpoint normalization

Proposition 10.2 (Real-parameter odd-zeta resolvent)**.** *For $k \geq 1$, $a > -1$, and $z = \sqrt{1 + a} > 0$,*

$$F_k(a) = z^{2k} \zeta(2k + 1, z) - \frac{1}{2z} - \frac{1}{2k}. \tag{10.12}$$

For $k = 0$,

$$F_0(a) = \log z - \frac{1}{2z} - \psi(z). \tag{10.13}$$

Proof. Binet's digamma formula [18] states that, for real $z > 0$,

$$h(z) := \log z - \frac{1}{2z} - \psi(z) = 2\int_0^\infty \frac{yB(2\pi y)}{y^2+z^2}\,\mathrm{d}y. \tag{10.14}$$

This immediately proves the case $k = 0$. For $k \geq 1$, use $\partial_y^{2k} B(2\pi y) = (2\pi)^{2k} B^{(2k)}(2\pi y)$ and integrate by parts $2k$ times:

$$F_k(z^2-1) = \frac{2}{(2k)!}\int_0^\infty B(2\pi y)\partial_y^{2k}\left(\frac{y^{2k+1}}{y^2+z^2}\right)\mathrm{d}y.$$

Every boundary term vanishes. At zero, the product of the relevant derivatives is $O(y)$: the rational factor has a zero of order $2k+1$, whereas $\partial_y^j B(2\pi y) = O(y^{-j-1})$. At infinity the exponential decay dominates the polynomial factors.

Polynomial division gives

$$\frac{y^{2k+1}}{y^2+z^2} = P_{2k-1}(y) + (-z^2)^k \frac{y}{y^2+z^2}.$$

The polynomial disappears after $2k$ differentiations. From $y/(y^2+z^2) = \frac{1}{2}((y+iz)^{-1}+(y-iz)^{-1})$ one obtains

$$\partial_y^{2k}\frac{y}{y^2+z^2} = (-1)^k\partial_z^{2k}\frac{y}{y^2+z^2}.$$

On compact positive z-intervals these integral derivatives are dominated near zero and at infinity. Consequently

$$F_k(z^2-1) = \frac{z^{2k}}{(2k)!}h^{(2k)}(z). \tag{10.15}$$

More explicitly, every fixed z-derivative of $y/(y^2+z^2)$ is $O(y)$ near $y = 0$, uniformly when z stays away from zero. Multiplication by $B(2\pi y) = O(y^{-1})$ therefore gives a locally integrable uniform bound. At infinity, the Bose factor supplies exponential domination. This verifies the exchange of differentiation and integration used in (10.15). Finally, $\psi^{(2k)}(z) = -(2k)!\zeta(2k+1,z)$ [19]; differentiating the logarithm and reciprocal in h supplies respectively $-1/(2k)$ and $-1/(2z)$ after multiplication by $z^{2k}/(2k)!$. This proves (10.12). □

Taking values and first derivatives at $a = 0$ gives

$$\begin{aligned} q_0^{(k)} &= \zeta(2k+1) - \frac{1}{2} - \frac{1}{2k}, \\ q_1^{(k)} &= \frac{2k+1}{2}\zeta(2k+2) - k\zeta(2k+1) - \frac{1}{4} \qquad (k \geq 1). \end{aligned} \tag{10.16}$$

For $k = 0$ the corresponding values are $q_0^{(0)} = \gamma - 1/2$ and $q_1^{(0)} = \zeta(2)/2 - 3/4$, where γ is Euler's constant. The differentiation uses $\partial_a = (2z)^{-1}\partial_z$ and $\partial_z\zeta(s,z) = -s\zeta(s+1,z)$. For $k \geq 1$, repetition expresses every fixed compact moment through finitely many adjacent zeta values and rational constants. The integral formulas guarantee, independently of any cancellation in those expressions, that $q_0^{(k)} > q_1^{(k)} > q_2^{(k)} > \cdots > 0$. This gives exact positivity identities suitable for comparing proposed moment constructions, without supplying the actual Θ jets.

The integer-parameter versions must retain their endpoint convention [14]. For $k \geq 1$, $N \geq 1$, and $s = 2k + 1$,

$$F_k(N^2 - 1) = N^{2k} \left(\zeta(s) - \sum_{n=1}^{N-1} n^{-s} \right) - \frac{1}{2N} - \frac{1}{2k}$$
$$= N^{2k} \left(\zeta(s) - \sum_{n=1}^{N} n^{-s} \right) + \frac{1}{2N} - \frac{1}{2k}. \tag{10.17}$$

Including the Nth summand changes the sign of the displayed $1/(2N)$ contribution. At $N = 1$, either version recovers the value of $q_0^{(k)}$ in (10.16).

10.4 An exact Bose term in the actual logarithmic derivative

There is a direct identity involving the actual completed zeta function. Let $a > 1/4$ and $s = 1/2 + \sqrt{a}$, so $s > 1$. Logarithmic differentiation of (1.2), Binet's formula, and the convergent von Mangoldt series [18, 20] give

$$2\sqrt{a}\, L(a) = \frac{1}{2s} + \frac{1}{s-1} + \frac{1}{2} \log \frac{s}{2\pi}$$
$$- \int_0^\infty \frac{y \, \mathrm{d}y}{(y^2 + s^2/4)(e^{2\pi y} - 1)} - \sum_{n=2}^{\infty} \Lambda(n) n^{-s}. \tag{10.18}$$

Indeed, before substitution the gamma contribution is $\frac{1}{2}\psi(s/2)$ and the elementary contribution includes $1/s$. The complete starting identity is

$$\frac{\xi'(s)}{\xi(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2}\psi(s/2) + \frac{\zeta'(s)}{\zeta(s)}.$$

Binet contributes $-1/(2s)$, leaving the coefficient $1/(2s)$ in (10.18). The same Bose kernel therefore occurs in the actual function, but together with the logarithmic, rational, and prime contributions. The positivity of one integral does not establish a positive-resolvent representation of their complete sum.

Proposition 10.3 (Asymptotic obstruction to direct calibration)**.** *No F_k equals the actual L on a nonempty open positive interval. More generally, this remains impossible for a finite sum $G(a) = \sum_{j=1}^{J} c_j F_{k_j}(d_j a + e_j)$ with $c_j, d_j > 0$ and $e_j \geq 0$.*

Proof. Dominated convergence in (10.6) gives

$$\lim_{a \to \infty} a F_k(a) = \frac{2k+1}{12}, \qquad \lim_{a \to \infty} a G(a) = \sum_{j=1}^{J} \frac{c_j (2k_j + 1)}{12 d_j} < \infty. \tag{10.19}$$

For the actual function, the integral in (10.18) is bounded by $1/(6s^2)$, using $\int_0^\infty y B(2\pi y)\, \mathrm{d}y = 1/24$. The prime sum is exponentially small as $s \to \infty$: for $s \geq 4$ it is bounded by $2^{-s/2} \sum_{n \geq 2} (\log n) n^{-2}$. The remaining rational terms are $O(s^{-1})$. Since $s = 1/2 + \sqrt{a}$, it follows that

$$L(a) = \frac{\log a}{8\sqrt{a}} - \frac{\log(2\pi)}{4\sqrt{a}} + O(a^{-1}), \qquad a L(a) \longrightarrow +\infty. \tag{10.20}$$

Both G and L are real analytic throughout $(0, \infty)$. Equality on an open subinterval would imply equality on that connected ray, contradicting (10.19) and (10.20). □

This obstruction is consistent with the abstract Hilbert-seed theorem. A finite compact spectral measure need not have a finite reciprocal moment. For the present models, however,

$$\int_{(0,1]} \frac{1}{t}\,\mathrm{d}\nu_k(t) = \int_0^\infty \omega_k(y)\,\mathrm{d}y = \frac{2k+1}{12}. \tag{10.21}$$

This extra integrability is exactly what makes $aF_k(a)$ tend to a finite limit. Under RH, the canonical measure in (8.3) instead has reciprocal moment $\sum_n 1 = \infty$, although its total mass is finite. Thus one cannot reject all bounded positive operators from Proposition 10.3. The result rejects this particular specific Bose multiplication construction and its stated finite combinations. It identifies a concrete property that a different construction would have to change.

Similarly, the positive matrices in (10.8) can serve as exact examples for checking normalization, shifts and proposed reconstruction lemmas. Their proofs work uniformly in degree because a specific positive measure was supplied first. Transferring that conclusion to the actual jets would require a theorem equating the entire moment sequences, not agreement of several low-order entries or a numerical fit on an interval.

The usable conclusion is precise: the hyperbolic identities supply explicit positive models, exact moment hierarchies, and a Bose integral inside the actual calibration problem. The bounded compactification above does not solve that problem. Any further construction must account for the full expression (10.18), including the growth in (10.20), before it can satisfy HSeed_Θ.

10.5 From hyperbolic source identities to the actual logarithmic moments

The useful connection begins with an identity for the actual completed function, rather than with a positive model chosen independently of it. Write $B(x) = (e^x - 1)^{-1}$ for $x > 0$. Elementary cancellation gives the positive density

$$h(x) := 4B(2x) - 8B(4x) = \frac{4}{e^{2x}+1} = 2(1 - \tanh x) > 0. \tag{10.22}$$

Thus a difference of two Bose kernels is exactly a positive Fermi, or hyperbolic-tangent, kernel. Its positivity follows after the cancellation; the two separate terms should not be treated as a positive-measure decomposition.

Define the Mellin integral

$$I(s) = \int_0^\infty x^{s-1} h(x)\,\mathrm{d}x = 2^{2-s}\Gamma(s)(1 - 2^{1-s})\zeta(s), \qquad \Re s > 0. \tag{10.23}$$

Here the expression at $s = 1$ is removable, with $I(1) = 2\log 2$. For $\Re s > 1$, expand $(e^{2x}+1)^{-1} = \sum_{k\ge1}(-1)^{k-1}e^{-2kx}$. Absolute convergence after integration permits termwise use of the gamma integral and gives $I(s) = 2^{2-s}\Gamma(s)\sum_{k\ge1}(-1)^{k-1}k^{-s}$. Separating even and odd terms proves (10.23) there. Both sides continue holomorphically to $\Re s > 0$: on the integral side, $h(x) = O(1)$ at zero and $h(x) = O(e^{-2x})$ at infinity; on the function side, the factor $1 - 2^{1-s}$ removes the pole of ζ at 1. The identity theorem gives the claimed domain. This is the scaled Fermi integral in [21, Eq. 25.5.3].

The same density is obtained from the explicitly defined two-variable measure

$$q(r) = \frac{r}{e^r - 1}, \qquad d(r) = -q'(r) > 0, \qquad \mathrm{d}\mu(r,y) = 2d(r)\,\mathrm{d}r\,\frac{\mathrm{d}y}{y}, \quad \frac{1}{4} \le y \le \frac{1}{2}, \tag{10.24}$$

by pushing forward under $x = ry$. Indeed its density is

$$2\int_{1/4}^{1/2} d(x/y)\,\frac{\mathrm{d}y}{y^2} = \frac{2}{x}(q(2x) - q(4x)) = h(x).$$

For completeness, $d(r) > 0$ because $-q'(r) = [e^r(r-1)+1]/(e^r-1)^2$, and the numerator vanishes at zero and has derivative $re^r > 0$. This calculation gives an exact one-variable realization of the source layer. In particular its total mass is $I(1) = 2\log 2$, and the original two-variable integral of $(ry)^{s-1}$ is precisely $I(s)$. Tonelli's theorem applies first to nonnegative integrands; absolute integrability then gives the same change of variables for the complex Mellin integrands used here.

The gamma duplication formula [22] simplifies the normalizing factor to

$$A(s) = \frac{s(s-1)\pi^{(1-s)/2}}{4\Gamma((s+1)/2)(1-2^{1-s})}, \qquad \xi(s) = A(s)I(s). \tag{10.25}$$

The equality holds in a complex neighborhood of $s = 1/2$, which is the domain needed here. The denominator is nonzero there, and A is positive on a real neighborhood of $1/2$: both $s(s-1)$ and $1-2^{1-s}$ are negative for $0 < s < 1$. We do not claim that the displayed A is holomorphic throughout the right half-plane; its other apparent singularities require cancellation in the product.

10.5.1 An explicit map from the source integrals to the required moments

Differentiation under the integral is valid to every fixed order:

$$I_j(s) := I^{(j)}(s) = \int_0^\infty x^{s-1}(\log x)^j h(x)\,\mathrm{d}x. \tag{10.26}$$

On each compact substrip $\Re s \geq \sigma > 0$, the behavior at zero is controlled by $x^{\sigma-1}|\log x|^j$, and the exponential tail controls infinity. In particular all the integrals at $1/2$ are absolutely convergent. The product rule now gives actual, unconditional coefficient identities:

$$a_n := \frac{\xi^{(2n)}(1/2)}{(2n)!\,\xi(1/2)}, \qquad a_0 = 1, \tag{10.27}$$

$$\xi^{(2n)}(1/2) = \sum_{j=0}^{2n} \binom{2n}{j} A^{(2n-j)}(1/2) I_j(1/2), \tag{10.28}$$

$$\frac{\Theta(z)}{\Theta(0)} = \sum_{n\geq 0} a_n z^n. \tag{10.29}$$

Consequently, hyperbolic identities that evaluate or bound these integrals can be transported to the actual Θ coefficients with every normalization factor visible.

The resolvent moments are different quantities. Set $m_n = (-1)^n L^{(n)}(0)/n!$ as before. Comparing coefficients in $\Theta' = L\Theta$ gives the exact triangular recurrence

$$(n+1)a_{n+1} = \sum_{k=0}^{n} (-1)^k m_k a_{n-k} \qquad (n \geq 0). \tag{10.30}$$

It determines every required m_n from $a_1, \ldots, a_{n+1}$. The first cases and the first nontrivial Hankel determinant are

$$\begin{aligned} m_0 &= a_1, \\ m_1 &= a_1^2 - 2a_2, \\ m_2 &= a_1^3 - 3a_1a_2 + 3a_3, \end{aligned} \tag{10.31}$$

$$\det \begin{pmatrix} m_0 & m_1 \\ m_1 & m_2 \end{pmatrix} = a_1^2 a_2 + 3a_1a_3 - 4a_2^2. \tag{10.32}$$

These are algebraic identities, not positivity assertions. In particular, positive formulas for the individual a_n do not settle the signs of their nonlinear combinations. To check the recurrence directly, multiply $\sum_{k\ge0}(-1)^k m_k z^k$ by $\sum_{j\ge0} a_j z^j$ in a sufficiently small disk where Θ is nonzero. The coefficient of z^n is the right side of (10.30), whereas differentiation of the second series gives its left side. The occurrence of m_n with coefficient $(-1)^n a_0 = (-1)^n$ makes the recursion unique.

There is also a compact cumulant interpretation. Let $X = \log x$ under the probability measure $\mathrm{d}\nu = x^{-1/2}h(x)\,\mathrm{d}x/I(1/2)$, and write $\mathrm{cum}_j(X)$ for its jth cumulant. Its moment generating function exists near zero, and $\mathrm{cum}_j(X) = (\log I)^{(j)}(1/2)$. Indeed $\mathbf{E}_\nu[e^{tX}] = I(1/2+t)/I(1/2)$ for $|\Re t| < 1/2$. The preceding endpoint estimates control this identity and all its fixed derivatives on smaller strips. Evenness of $\xi(1/2+w)$ and its nonzero value at zero give

$$m_n = \frac{(-1)^n}{2(2n+1)!}\Big((\log A)^{(2n+2)}(1/2) + \mathrm{cum}_{2n+2}(X)\Big). \tag{10.33}$$

All logarithms here are local analytic logarithms chosen real on the relevant real interval. Formula (10.33) shows explicitly why ordinary positive-measure moments do not automatically yield the required alternating logarithmic moments. The derivatives of $\log A$ are part of the calibration and cannot be discarded. For example, the second derivative of $\log I$ is the variance of X, but the second derivative of $\log\xi$ also contains $(\log A)''$. Higher cumulants have no positivity law comparable to the elementary squared-norm identity for ordinary moment matrices.

10.5.2 The positive mass and the remaining positivity theorem

The mass $\kappa = m_0$ is strictly positive without RH. One standard positive Riemann kernel, obtained by rescaling the representation in [23, Section 4, Eqs. (4.1)–(4.2)], is

$$\Psi(u) = 4\sum_{n\ge1}\pi n^2\big(2\pi n^2 e^{2u} - 3\big)e^{5u/2-\pi n^2 e^{2u}}, \qquad u \ge 0, \tag{10.34}$$

$$\xi(1/2+w) = \int_0^\infty \Psi(u)\cosh(wu)\,\mathrm{d}u. \tag{10.35}$$

Every summand is positive, since $2\pi n^2 e^{2u} > 3$. The rapid decay justifies differentiation, and therefore

$$\kappa = \frac{\int_0^\infty u^2\Psi(u)\,\mathrm{d}u}{2\int_0^\infty \Psi(u)\,\mathrm{d}u} > 0. \tag{10.36}$$

To see the distinction quantitatively, normalize $\mathrm{d}\nu_R = \Psi(u)\,\mathrm{d}u/\xi(1/2)$ and put $b_j = \int u^{2j}\,\mathrm{d}\nu_R$, so that $a_j = b_j/(2j)!$ and $b_0 = 1$. Then

$$m_1 = \frac{3b_1^2 - b_2}{12}. \tag{10.37}$$

Ordinary positive-measure Cauchy–Schwarz gives $b_2 \ge b_1^2$; the necessary resolvent inequality $m_1 \ge 0$ instead requires the additional upper bound $b_2 \le 3b_1^2$. These are different inequalities. The first Hankel determinant in (10.32) introduces still another combination involving b_3, and the hierarchy continues to unbounded degree.

Such kernel positivity must still be distinguished from the full resolvent criterion. For example, the entire function $\Theta_*(z) = \frac{3}{4} + \frac{1}{4}\cosh\sqrt{z}$ has a positive cosh-measure representation and is positive for $z \ge 0$. Nevertheless its normalized coefficients satisfy $a_1 = 1/8$, $a_2 = 1/96$, and its transformed moment is $m_1 = a_1^2 - 2a_2 = -1/192 < 0$. Thus even a positive cosh representation and entire square descent do not by themselves imply the desired Stieltjes moment positivity.

The exact benefit of the hyperbolic source identities is therefore the calibrated chain $I_j \longmapsto \xi^{(2n)}(1/2) \longmapsto a_n \longmapsto m_n$. The missing theorem must control these *actual transformed* moments:

for one finite $M > 0$, all finite sections of (m_{i+j}), (m_{i+j+1}), and $(M^2 m_{i+j} - m_{i+j+2})$ must be positive semidefinite. The source integrals already give positivity of ordinary log-moment matrices $(I_{i+j}(1/2))$, since for every finite coefficient list

$$\sum_{i,j} \overline{c}_i c_j I_{i+j}(1/2) = \int_0^\infty x^{-1/2} h(x) \left| \sum_j c_j (\log x)^j \right|^2 \mathrm{d}x \geq 0.$$

These are moments of $\log x$, which ranges over the whole real line. They are not already moments of a bounded nonnegative operator with the required logarithmic-derivative calibration. Transferring their positivity through (10.28) and (10.30) is a substantive additional obligation. Establishing it uniformly would activate the paper's bounded moment reconstruction and RH transfer.

10.6 From the signed hyperbolic pencil to positive determinant models

The determinant identities in the Hyperbolic Zeta project give a second construction, different from the multiplication model above. Let k, m, N, J be integers with $k, m \geq 1$ and $1 \leq N < J$, and put $h = J - N$, $x_0 = \zeta(2k+1)$ and

$$D_j(t) = \prod_{r=1}^{j} (t + r^2).$$

For an admissible affine Hankel pencil in that project, its value G at x_0 has entries

$$G_{ij} = \int_0^\infty y^{2(i+j)} \frac{D_N(y^2)^{2m}}{D_J(y^2)} \omega_k(y)\, \mathrm{d}y, \qquad 0 \leq i, j < h. \tag{10.38}$$

All these integrals converge. For $c \in \mathbb{C}^h \setminus \{0\}$,

$$c^* G c = \int_0^\infty \left| \sum_{j=0}^{h-1} c_j y^{2j} \right|^2 \frac{D_N(y^2)^{2m}}{D_J(y^2)} \omega_k(y)\, \mathrm{d}y > 0.$$

The density and rational factor are positive, and a nonzero polynomial cannot vanish on an interval. Thus G is positive definite. The affine pencil is

$$\mathcal{G}(X) = G + (X - x_0)M, \qquad \Delta(X) = \det \mathcal{G}(X), \qquad B = G^{-1/2} M G^{-1/2}. \tag{10.39}$$

Here $M = M^*$ is the coefficient of X. These identifications are recorded in the project's exact Gram and pole-diagonalization sources [16, 17].

The positive Gram matrix does not make the coefficient M positive. In the pole basis the identity is

$$M = V \operatorname{diag}(\gamma_r)_{N<r\leq J} V^T, \qquad \operatorname{sgn} \gamma_r = (-1)^{r-1}, \tag{10.40}$$

with V invertible. In fact,

$$\gamma_r = r^{2k} \frac{D_N(-r^2)^{2m-1}}{D'_{\mathrm{tail}}(-r^2)}, \qquad D_{\mathrm{tail}}(t) = \prod_{j=N+1}^{J} (t + j^2).$$

The numerator has sign $(-1)^N$ and the denominator has sign $(-1)^{r-N-1}$, proving the sign in (10.40). Consequently B is self-adjoint and invertible, but has both positive and negative eigenvalues when $h \geq 2$. Hyperbolicity of Δ means its roots are real; the Hilbert seed instead requires a nonnegative operator. These are different properties.

Proposition 10.4 (Squared-pencil identity)**.** *For any finite self-adjoint B as in (10.39) and any $c > 0$, let $K = cB^2$ and $Q_c(z) = \det(I + zK)$. Then*

$$Q_c(z) = \frac{\Delta(x_0 + i\sqrt{cz})\Delta(x_0 - i\sqrt{cz})}{\Delta(x_0)^2}. \tag{10.41}$$

The right side is a polynomial in z, independent of the choice of square root. If $\beta_1, \dots, \beta_h$ are the eigenvalues of B, then $\lambda_j = c\beta_j^2 \geq 0$ and

$$Q_c(z) = \prod_{j=1}^{h}(1 + z\lambda_j), \qquad \frac{Q_c'(a)}{Q_c(a)} = \sum_{j=1}^{h}\frac{\lambda_j}{1 + a\lambda_j} = \langle v, (I + aK)^{-1}v\rangle, \quad a \geq 0, \tag{10.42}$$

where $v = \sum_j \sqrt{\lambda_j}\, e_j$ in an orthonormal eigenbasis of K. In particular the exact differential seed algebra holds for Q_c'/Q_c.

Proof. Congruence in (10.39) gives $\Delta(x_0 + u)/\Delta(x_0) = \det(I + uB)$. The commuting factors satisfy

$$(I + i\sqrt{cz}B)(I - i\sqrt{cz}B) = I + czB^2.$$

Taking determinants proves (10.41). The spectral theorem gives the product and the scalar resolvent in (10.42). Applying the already proved identity (3.8) to this K, v gives the differential identity. For real $z \geq 0$, the numerator in (10.41) is an absolute square; for general complex z it is the product of the two displayed factors. □

The operation $B \mapsto B^2$ changes the scalar determinant. It therefore constructs positive candidates without transferring any identity with the actual Θ. The next result states a sufficient transfer condition in terms of exactly specified coefficients.

10.7 An all-order coefficient-limit theorem

Write $\Theta(z)/\Theta(0) = \sum_{r\geq 0} a_r z^r$, where $a_0 = 1$, and let $\kappa = a_1 = L(0) > 0$ as established above. Consider a sequence of finite, nonzero self-adjoint matrices B_n, for example a specified sequence of the hyperbolic slopes. Define

$$c_n = \frac{\kappa}{\operatorname{Tr}(B_n^2)}, \qquad K_n = c_n B_n^2, \qquad Q_n(z) = \det(I + zK_n) = \sum_{r\geq 0} e_{n,r} z^r. \tag{10.43}$$

Coefficients above the degree of Q_n are set equal to zero. The denominator is positive, and

$$K_n \geq 0, \qquad \operatorname{Tr} K_n = \kappa, \qquad \|K_n\| \leq \kappa, \qquad 0 \leq e_{n,r} \leq \frac{\kappa^r}{r!}. \tag{10.44}$$

The last inequality follows by expanding $(\sum_j \lambda_{n,j})^r$: the terms with r distinct indices include $r!$ copies of every summand of the elementary symmetric polynomial $e_{n,r}$.

Theorem 10.5 (Conditional transfer of hyperbolic determinant models)**.** *For the sequence (10.43), the following two conditions are equivalent:*

$$\text{for every } r \geq 0, \qquad \lim_{n\to\infty} e_{n,r} = \frac{\xi^{(2r)}(1/2)}{(2r)!\,\xi(1/2)} = a_r; \tag{10.45}$$

$$\text{for every } r \geq 0, \qquad \lim_{n\to\infty} \operatorname{Tr}(K_n^{r+1}) = \frac{(-1)^r}{r!}L^{(r)}(0) = m_r. \tag{10.46}$$

Either condition implies locally uniform convergence on $\mathbb{C}$,

$$Q_n(z) \longrightarrow \frac{\Theta(z)}{\Theta(0)}, \tag{10.47}$$

and implies RH. No assertion that the particular Hyperbolic Zeta sequence satisfies either condition is included in this theorem.

Proof. Put $p_{n,j} = \operatorname{Tr}(K_n^j)$ for $j \geq 1$. Newton's identities give

$$re_{n,r} = \sum_{j=1}^{r} (-1)^{j-1} e_{n,r-j} p_{n,j}.$$

The actual coefficients satisfy the identical finite algebraic recurrence

$$ra_r = \sum_{j=1}^{r} (-1)^{j-1} a_{r-j} m_{j-1},$$

obtained from $\Theta' = L\Theta$. For any fixed r, the first recurrence determines $e_{n,r}$ from $p_{n,1}, \ldots, p_{n,r}$ and also determines $p_{n,r}$ from $e_{n,1}, \ldots, e_{n,r}$, since $e_{n,0} = 1$. Induction on r proves the equivalence of (10.45) and (10.46).

Assume coefficient convergence. Taking limits in (10.44) gives $0 \leq a_r \leq \kappa^r / r!$. On every disk $|z| \leq R$, both coefficient series are dominated by $\sum_r (\kappa R)^r / r! = e^{\kappa R}$. Split the series at a fixed degree, use coefficient convergence for its finite head, and then let that degree tend to infinity. This proves (10.47). Quantitatively, for every integer $d \geq 0$,

$$\sup_{|z| \leq R} \left| Q_n(z) - \frac{\Theta(z)}{\Theta(0)} \right| \leq \sum_{r=0}^{d} |e_{n,r} - a_r| R^r + \frac{2e^{\kappa R} (\kappa R)^{d+1}}{(d+1)!}. \tag{10.48}$$

Every zero of Q_n is $-1/\lambda_{n,j}$ for a positive eigenvalue. Since $\lambda_{n,j} \leq \kappa$, all Q_n are zero-free on the connected domain $\mathbb{C} \setminus (-\infty, -1/\kappa]$. Their locally uniform limit equals 1 at the origin and is not identically zero. Hurwitz's theorem therefore makes this limit zero-free on the same domain. The zeros of Θ lie on the negative real axis, giving RH by Section 2.

There is also a direct route through the moment criterion. The measures $\mu_n = \sum_j \lambda_{n,j} \delta_{\lambda_{n,j}}$ have mass κ and support in $[0, \kappa]$. Hence their moments satisfy all three quadratic-form inequalities in Section 8, with $M = \kappa$. Under (10.46), taking limits in each finite quadratic form transfers those inequalities to the actual m_r. The reconstruction theorem then supplies the calibrated seed. □

The fixed Hilbert space in this limit can be displayed directly. The measures μ_n in the proof have uniformly bounded mass on the compact interval $[0, \kappa]$, so they have weakly convergent subsequences. Condition (10.46) fixes every moment of any such limit. Polynomials are dense in the continuous functions on this interval, so the limit measure is unique; call it μ. Then

$$\mathcal{H} = L^2(\mu), \qquad (Kf)(t) = tf(t), \qquad v(t) = 1 \tag{10.49}$$

satisfy $0 \leq K \leq \kappa I$ and $\|v\|^2 = \kappa$. The Taylor series of $\langle v, (I + zK)^{-1} v \rangle = \int (1 + zt)^{-1} \, \mathrm{d}\mu(t)$ has coefficients $(-1)^r m_r$, so it equals the actual L near zero. Thus the conditional limit constructs one fixed seed of the type used in the main theorem.

For the squared pencils, the first two nonautomatic trace targets are

$$\lim_{n \to \infty} \frac{\operatorname{Tr} B_n^4}{(\operatorname{Tr} B_n^2)^2} = \frac{m_1}{\kappa^2}, \qquad \lim_{n \to \infty} \frac{\operatorname{Tr} B_n^6}{(\operatorname{Tr} B_n^2)^3} = \frac{m_2}{\kappa^3}. \tag{10.50}$$

These can serve as preliminary consistency tests for a proposed choice of pencil parameters. They become an RH argument only when the all-order statement (10.46) is proved.

Example 10.6 (Trace normalization does not give calibration)**.** Take $K_n = (\kappa/n) I_n$, which has nonnegative spectrum and trace κ. Then

$$Q_n(z) = (1 + \kappa z/n)^n \longrightarrow e^{\kappa z}, \qquad \operatorname{Tr}(K_n^{r+1}) = \kappa^{r+1} n^{-r}.$$

This limit has no zeros, whereas the actual Θ has infinitely many zeros. Thus positivity, growing dimension, exact normalization, and locally uniform convergence do not by themselves identify the limit with Θ. The example is a general normalization test, not an assertion that the project's particular pencils have this limit.

The remaining Hyperbolic Zeta obligation. Choose an explicit family of admissible pencils and prove (10.45), equivalently (10.46), for every degree. The constructions and identities above establish a positive candidate family and the exact consequence of such a limit theorem. They leave that limit theorem open. The normalization uses the actual real constant κ, and the pencil data use actual zeta values. Finite matrix dimension therefore does not make this construction a finite raw checker witness. The implication proved here is convergence of this family implies RH; a converse for this particular family would require a separate completeness theorem.

11 Canonical actual polynomials and an explicit extension problem

The arbitrary squared-pencil family of Section 10.7 has an open coefficient-identification obligation. A different construction starts directly from the actual coefficients in (10.27). For this canonical family, coefficient convergence is unconditional; the remaining requirement is real-rootedness. This is the classical Jensen–Pólya criterion, specialized to the present normalization and connected explicitly to the Hilbert seed. No novelty of that criterion is claimed; see [24, 25].

11.1 Actual coefficients determine the approximants

Write

$$T(z) = \frac{\Theta(z)}{\Theta(0)} = \sum_{r\ge 0} a_r z^r, \qquad a_0 = 1, \quad a_1 = \kappa.$$

The positive kernel in (10.35) shows that $a_r > 0$ for every r: each even derivative is the integral of $u^{2r}\Psi(u)$, and the density is strictly positive on $(0,\infty)$. For $n \ge 1$ define the falling factorial $(n)_r = n(n-1)\cdots(n-r+1)$, with $(n)_0 = 1$, and set

$$P_n(z) = \sum_{r=0}^{n} c_{n,r} z^r, \qquad c_{n,r} = \frac{(n)_r}{n^r} a_r. \tag{11.1}$$

Set $c_{n,r} = 0$ for $r > n$. These are degree-n polynomials with $P_n(0) = 1$, $P_n'(0) = \kappa$, and strictly positive coefficients. In the usual Jensen notation for $\widehat{\gamma}_r = r! a_r$, $P_n(z) = J^{n,0}_{\widehat{\gamma}}(z/n)$.

Proposition 11.1 (Unconditional calibration of the polynomial limit)**.** *For every $R \ge 0$ and $n \ge 1$,*

$$\sup_{|z|\le R} |T(z) - P_n(z)| \le \frac{R^2 T''(R)}{2n}. \tag{11.2}$$

Consequently $P_n \to T$ locally uniformly on $\mathbb{C}$, without assuming RH or real-rootedness of any P_n.

Proof. For $0 \le r \le n$, the elementary product inequality $1 - \prod_j (1 - x_j) \le \sum_j x_j$ for $0 \le x_j \le 1$ gives

$$0 \le 1 - \frac{(n)_r}{n^r} \le \frac{r(r-1)}{2n}.$$

For $r > n$, the omitted coefficient has multiplier difference 1, which is also at most $r(r-1)/(2n)$. Multiplication by $a_r R^r$ and summation over r proves (11.2), since T is entire and its coefficients are positive. □

The logarithmic derivatives are calibrated as limits as well. Since $P_n(a), T(a) \ge 1$ for $a \ge 0$, differentiating the coefficient estimate yields

$$\sup_{0\le a\le R} \left| \frac{P_n'(a)}{P_n(a)} - L(a) \right| \le \frac{R^2 T'''(R) + 2RT''(R) + R^2 T'(R) T''(R)}{2n}. \tag{11.3}$$

Indeed, the bound for $T'-P_n'$ is $[R^2T'''(R)+2RT''(R)]/(2n)$, and $P_n'/P_n-T'/T=(P_n'-T')/P_n+T'(T-P_n)/(P_nT)$. This proves (11.3). Neither estimate asserts that these rational logarithmic derivatives have a positive-resolvent representation.

11.2 Real-rootedness supplies the positive matrices

Lemma 11.2. *If a nonzero real polynomial q has all its zeros in $(-\infty,0]$ and $\lambda\ge 0$, then $q+\lambda q'$ has the same property.*

Proof. The cases $\lambda=0$ and constant q are immediate. For $\lambda>0$, let the distinct zeros of q be $r_1<\cdots<r_d\le 0$ with multiplicities b_i. Away from them, $q'/q=\sum_i b_i/(x-r_i)$ is strictly decreasing on each gap. The equation $q'/q=-1/\lambda$ has one solution in $(-\infty,r_1)$ and one in each (r_i,r_{i+1}), and none to the right of r_d. Each r_i remains a zero of multiplicity b_i-1. These account for all $\deg q$ roots. □

Theorem 11.3 (Canonical polynomial criterion)**.** *The following statements are equivalent:*

(i) RH holds.

(ii) Every P_n, $n\ge 1$, has only real zeros.

(iii) There are arbitrarily large n for which P_n has only real zeros.

Whenever P_n has only real zeros, they are strictly negative and there is a positive diagonal matrix K_n^{J} such that

$$P_n(z)=\det(I+zK_n^{\mathrm{J}}),\qquad \operatorname{Tr}K_n^{\mathrm{J}}=\kappa,\qquad 0<K_n^{\mathrm{J}}\le\kappa I. \tag{11.4}$$

Proof. Assume RH. By (7.3), the finite products $q_s(z)=\prod_{j=1}^{s}(1+\lambda_j z)$, with $\lambda_j=\gamma_j^{-2}>0$, converge locally uniformly to T. For fixed n, apply $q_s(D)$ to x^n, where $D=d/dx$. Lemma 11.2, applied successively to the factors $1+\lambda_jD$, makes $q_s(D)x^n$ real-rooted with nonpositive zeros. Its reciprocal polynomial is

$$z^n\,[q_s(D)x^n]_{x=1/z}=\sum_{r=0}^{n}(n)_r[z^r]q_s(z)\,z^r.$$

The expression is understood as a polynomial, including any cancellation caused by zeros at $x=0$. Its finite zeros are negative. Scale z by $1/n$ and let $s\to\infty$. Coefficients converge to those of P_n, whose degree is n and constant term is 1. Real-rootedness is closed under such coefficient limits: a nonreal zero of the limit would, by Rouché's theorem on a small disk avoiding the real axis, force a nonreal zero in the approximants. Thus P_n has only real zeros. Its positive coefficients and nonzero constant term exclude nonnegative zeros.

For any such polynomial, factor $P_n(z)=\prod_{j=1}^{n}(1+\lambda_{n,j}z)$ with $\lambda_{n,j}>0$. Its linear coefficient gives $\sum_j\lambda_{n,j}=\kappa$, proving (11.4). Thus every such P_n is zero-free on $D_\kappa=\mathbb{C}\setminus(-\infty,-1/\kappa]$. If (iii) holds, Proposition 11.1 and Hurwitz's theorem show that T is zero-free on D_κ, since $T(0)=1$. All zeros of Θ are therefore negative real, which is RH by Section 2. The implication (ii)⇒(iii) is immediate. □

For a hyperbolic P_n, take $(v_n)_j=\sqrt{\lambda_{n,j}}$. Then

$$\frac{P_n'(a)}{P_n(a)}=\langle v_n,(I+aK_n^{\mathrm{J}})^{-1}v_n\rangle,\qquad \|v_n\|^2=\kappa. \tag{11.5}$$

Newton's identities and $c_{n,r}\to a_r$ give, for each fixed r,

$$\operatorname{Tr}\big((K_n^{\mathrm{J}})^{r+1}\big)\longrightarrow m_r \tag{11.6}$$

along any unbounded sequence of hyperbolic degrees. The moment reconstruction in Section 8 then supplies one fixed seed. The first two nonconstant trace identities, when the positive matrix exists, are

$$\operatorname{Tr}((K_n^{\mathrm{J}})^2) = m_1 + \frac{2a_2}{n}, \tag{11.7}$$

$$\operatorname{Tr}((K_n^{\mathrm{J}})^3) = m_2 + \frac{3\kappa a_2 - 9a_3}{n} + \frac{6a_3}{n^2}. \tag{11.8}$$

For the respective formulas take $n \geq 2$ and $n \geq 3$. They follow from $c_{n,2} = (1 - 1/n)a_2$, $c_{n,3} = (1 - 3/n + 2/n^2)a_3$ and the first Newton identities. The corresponding algebraic power sums exist even when the roots are not real; positivity of their representing matrix is the unresolved condition in that case.

11.3 A finite matrix test involving only actual coefficients

The roots can be removed from the formulation. Put

$$C_n(t) = t^n P_n(-1/t) = \sum_{r=0}^{n} (-1)^r c_{n,r} t^{n-r}. \tag{11.9}$$

Let $s_{n,0} = n$ and calculate $s_{n,j}$ by Newton's identities:

$$s_{n,j} + \sum_{r=1}^{j-1} (-1)^r c_{n,r} s_{n,j-r} + (-1)^j j c_{n,j} = 0 \qquad (1 \leq j \leq n), \tag{11.10}$$

$$s_{n,j} + \sum_{r=1}^{n} (-1)^r c_{n,r} s_{n,j-r} = 0 \qquad (j > n). \tag{11.11}$$

Define the real symmetric Hermite matrix

$$\mathcal{E}_n = (s_{n,i+j})_{0 \leq i,j < n}. \tag{11.12}$$

These are explicit algebraic expressions in $a_1, \ldots, a_n$. The classical Hermite–Sylvester criterion gives

$$\mathcal{E}_n \succeq 0 \quad \Longleftrightarrow \quad P_n \text{ is hyperbolic.} \tag{11.13}$$

For completeness, if the roots of C_n are λ_j counted with multiplicity, its quadratic form at the coefficients of a real polynomial p of degree below n is $\sum_j p(\lambda_j)^2$. This is nonnegative when all roots are real. If a conjugate nonreal pair exists, real Lagrange interpolation assigns values $i, -i$ to that pair and zero to the other distinct roots. The quadratic form is then $-2b < 0$, where b is the pair's multiplicity. This proves (11.13), including repeated roots; see [26, Theorem 1].

Combining with Theorem 11.3 yields

$$\text{RH} \quad \Longleftrightarrow \quad \mathcal{E}_n \succeq 0 \text{ for every } n \geq 1 \quad \Longleftrightarrow \quad \mathcal{E}_n \succeq 0 \text{ for arbitrarily large } n. \tag{11.14}$$

This is a concrete finite matrix at each degree. It still has an unbounded degree quantifier, and its entries contain the actual analytic constants. It is not yet the finite raw-data checker of (15.1).

11.4 Exact hyperbolic source formula and a literature base

Let $s_0 = 1/2$, $h(t) = 4/(e^{2t} + 1)$, and retain the complete prefactor $A(s)$ from (10.25). Define

$$\mathcal{K}_n(z, y) = \sum_{r=0}^{n} \frac{(n)_r z^r}{n^r (2r)!} \sum_{j=0}^{2r} \binom{2r}{j} A^{(2r-j)}(s_0) y^j. \tag{11.15}$$

Substituting (10.28) into (11.1) gives the exact finite-integral identity

$$P_n(z) = \frac{1}{\xi(s_0)} \int_0^\infty t^{-1/2} h(t)\, \mathcal{K}_n(z, \log t)\, \mathrm{d}t. \tag{11.16}$$

All interchanges here are finite sums of the absolutely convergent logarithmic moments already proved in Section 10.5. Thus the hyperbolic source determines every coefficient of the canonical polynomial and every entry of (11.12). Positivity of h does not supply the matrix inequality $\mathcal{E}_n \succeq 0$: its entries involve Newton transformations and the full derivatives of A.

There is an established finite base for this family. Griffin–Ono–Rolen–Zagier prove that their Jensen polynomials are hyperbolic for degrees $1 \le d \le 8$ at every nonnegative shift [24, Theorem 2]. Their coefficient convention is $\gamma(r) = 8\xi(1/2)\, r! a_r$, so

$$P_n(z) = \frac{J_\gamma^{n,0}(z/n)}{8\xi(1/2)}.$$

Their theorem therefore supplies hyperbolicity of our $P_1, \dots, P_8$. This is a cited classical base, not a fresh computation or native Lean certificate, and degree eight is not claimed to be the largest known range.

The effective large-shift theorem in [25, Theorem 1.1] has different quantifiers: for each degree d, the shift can be taken sufficiently large. The present approximants keep the shift equal to zero and let the degree grow. A result about large shifts does not establish the unbounded-degree, zero-shift condition in (11.14).

Exact change in the remaining obligation. For the canonical family (11.1), its coefficients, entire limit, and scalar logarithmic-derivative limit are now specified and calibrated unconditionally. A fixed positive Hilbert seed follows if hyperbolicity, equivalently the actual matrices $\mathcal{E}_n \succeq 0$, is established at unbounded degrees. The paper does not prove that all-degree positivity assertion. This construction supplements the original Hyperbolic Zeta pencil route; it does not prove the latter's coefficient limits or its completeness.

11.5 The exact upward extension condition

The all-degree criterion has a precise one-degree extension condition. For $a_0 = 1$, define the monic polynomials

$$H_n(X) = \sum_{r=0}^{n} (n)_r a_r X^{n-r}, \qquad (n)_r = \frac{n!}{(n-r)!}. \tag{11.17}$$

They are the reciprocal form of the canonical approximants: $P_n(z) = (z/n)^n H_n(n/z)$, interpreted by polynomial continuation at $z = 0$. Thus P_n is hyperbolic if and only if H_n is hyperbolic. The initial polynomial $H_1(X) = X + \kappa$ is already real-rooted. Direct differentiation gives

$$H_n' = nH_{n-1}, \qquad H_n(X) = R_n(X) + n! a_n, \qquad R_n(X) = n \int_0^X H_{n-1}(t)\, \mathrm{d}t. \tag{11.18}$$

Thus the preceding polynomial determines the derivative, while the actual next coefficient determines the remaining additive constant.

Lemma 11.4 (Critical-value test, including multiplicities)**.** *Let P be a real monic polynomial of degree $n \ge 1$. Suppose P'/n has only real roots, listed with multiplicity as $c_1 \le \cdots \le c_{n-1}$. Then P has only real roots if and only if*

$$(-1)^{n-i} P(c_i) \ge 0 \qquad (1 \le i \le n-1). \tag{11.19}$$

For $n = 1$, the condition is empty.

Proof. If P is real-rooted, write its roots as $x_1 \le \cdots \le x_n$. Rolle's theorem, including the inherited multiplicities at repeated roots, gives $x_i \le c_i \le x_{i+1}$. A monic polynomial has sign $(-1)^{n-i}$ between these roots, with zero allowed at an endpoint. This proves necessity.

Conversely assume (11.19). Group the distinct critical points, and let m be the multiplicity of one of them as a root of P'. If $m \ge 2$, two adjacent inequalities have opposite signs and the same value $P(c)$, forcing $P(c) = 0$. Whenever $P(c) = 0$, Taylor expansion shows that its multiplicity as a root of P is exactly $m+1$. Consequently every critical point with a nonzero value is simple. Two zero-valued critical points cannot be consecutive distinct critical points: between them P' would have a constant strict sign, contradicting equality of the endpoint values.

Let z count the distinct zero-valued critical points, q the nonzero-valued critical points, and $M = \sum_{P(c)=0}(m_c+1)$. Counting the roots of P' gives $\sum_{P(c)=0} m_c + q = n-1$, hence $n - M = q+1-z$. There are $q+z+1$ open gaps between consecutive distinct critical points and the two infinities. Exactly $2z$ meet a zero-valued critical point, since no two such points are consecutive. Each of the remaining $q+1-z$ gaps has strictly opposite signs at its ends: this follows from (11.19) at adjacent nonzero critical points, and from the monic leading term at infinity. The intermediate value theorem supplies one real root in each gap. Together with the M roots counted with multiplicity at critical points, this supplies n real roots. The degree is n, proving sufficiency. □

If H_{n-1} is real-rooted, Lemma 11.4 turns its next step into the exact scalar interval

$$\max_{\substack{1\le i<n\\ n-i \text{ even}}} \{-R_n(c_i)\} \;\le\; n!a_n \;\le\; \min_{\substack{1\le i<n\\ n-i \text{ odd}}} \{-R_n(c_i)\}. \tag{11.20}$$

An empty maximum is $-\infty$, and an empty minimum is $+\infty$. In particular, the degree-two condition has only an upper bound. Repeated critical roots require no perturbation: adjacent opposite inequalities enforce the vanishing and multiplicity used in the proof.

11.5.1 The first gates in actual logarithmic moments

The actual moments make the first conditions particularly explicit. The generating identity, interpreted near $t = 0$, is

$$\begin{aligned}\sum_{n\ge 0} H_n(y-\kappa)\frac{t^n}{n!} &= e^{(y-\kappa)t}\frac{\Theta(t)}{\Theta(0)}\\ &= \exp\left(yt - \frac{m_1t^2}{2} + \frac{m_2t^3}{3} - \frac{m_3t^4}{4} + \cdots\right).\end{aligned} \tag{11.21}$$

Here the local logarithm is legitimate because $\Theta(0) \ne 0$; no global zero-free logarithm is assumed. Coefficient comparison yields

$$\begin{aligned}H_2(y-\kappa) &= y^2 - m_1,\\ H_3(y-\kappa) &= y^3 - 3m_1y + 2m_2,\\ H_4(y-\kappa) &= y^4 - 6m_1y^2 + 8m_2y + 3m_1^2 - 6m_3.\end{aligned} \tag{11.22}$$

Thus H_2 is real-rooted exactly when $m_1 \ge 0$, equivalently $2a_2 \le a_1^2$. Given this condition, H_3 is real-rooted exactly when

$$m_2^2 \le m_1^3. \tag{11.23}$$

For $m_1 > 0$, evaluate the cubic at its critical points $y = \pm\sqrt{m_1}$ and apply Lemma 11.4. For $m_1 = 0$, the repeated critical point forces $m_2 = 0$.

Assume the cubic gate and list the roots of $y^3 - 3m_1y + 2m_2$ as $r_1 \le r_2 \le r_3$, with multiplicity. Set

$$g(r) = \frac{m_1^2 - m_1 r^2 + 2m_2 r}{2},$$
$$H_4 \text{ is real-rooted} \iff \max\{g(r_1), g(r_3)\} \le m_3 \le g(r_2). \tag{11.24}$$

Indeed the cubic equation reduces $H_4(r_i - \kappa)$ to $6(g(r_i) - m_3)$, and the required signs are respectively nonpositive, nonnegative, and nonpositive.

These formulas are exact extension tests. They do not establish their inequalities for the prescribed actual coefficients in every degree. The hyperbolic integral identities determine those coefficients before the test is applied; choosing a different admissible additive constant would construct a different function. A successful upward induction must prove (11.20) for the actual a_n at every stage. The unresolved issue is this uniform actual-coefficient theorem, not the existence of the polynomial derivative identity.

12 Canonical grid seeds from actual moments

There is a second canonical construction in which the actual logarithmic moments determine every finite model. It specializes the classical Hausdorff–Bernstein moment mechanism to this problem; general moment background is discussed in [27]. The proof below is self-contained relative to the reconstruction in Section 8. The hyperbolic source identities of Section 10.5 supply the required actual numbers. Their all-level positivity remains a separate assertion.

12.1 Explicit weights and unconditional moment convergence

Retain the actual moments $m_r = (-1)^r L^{(r)}(0)/r!$ and the positive constant $\kappa = m_0$. Define

$$u_r = \frac{m_r}{\kappa^{r+1}}, \qquad u_0 = 1, \qquad \ell(t^r) = u_r, \tag{12.1}$$

where ℓ is extended linearly to polynomials. For $0 \le k \le N$, put

$$b_{N,k}(t) = \binom{N}{k} t^k (1-t)^{N-k},$$
$$w_{N,k} := \ell(b_{N,k}) = \binom{N}{k} \sum_{j=0}^{N-k} (-1)^j \binom{N-k}{j} u_{k+j}. \tag{12.2}$$

These expressions use only $u_0, \dots, u_N$; their definition assumes no sign condition. The binomial identities give

$$\sum_{k=0}^{N} w_{N,k} = 1, \qquad \sum_{k=0}^{N} \frac{k}{N} w_{N,k} = u_1 \quad (N \ge 1). \tag{12.3}$$

Write $(q)_{\underline{j}} = q(q-1)\cdots(q-j+1)$, with $(q)_{\underline{0}} = 1$. The polynomial identity

$$\sum_{k=0}^{N} (k)_{\underline{j}} b_{N,k}(t) = (N)_{\underline{j}} t^j \qquad (0 \le j \le N)$$

follows from the coefficient relation

$$(k)_{\underline{j}} \binom{N}{k} = (N)_{\underline{j}} \binom{N-j}{k-j}.$$

Applying ℓ, and then the defining identity $k^r = \sum_{j=0}^{r} S(r,j)(k)_{\underline{j}}$ for Stirling numbers of the second kind, proves

$$\beta_{N,r} := \sum_{k=0}^{N} \left(\frac{k}{N}\right)^r w_{N,k} = \frac{1}{N^r} \sum_{j=0}^{\min(r,N)} S(r,j)(N)_{\underline{j}} u_j \quad (N \geq 1). \tag{12.4}$$

For $r = 0$ the monomial is the constant 1. In particular,

$$\beta_{N,0} = 1, \qquad \beta_{N,1} = u_1, \qquad \beta_{N,2} = u_2 + \frac{u_1 - u_2}{N}. \tag{12.5}$$

For every fixed r, the coefficient of u_r in (12.4) tends to 1 and the other coefficients tend to zero. Therefore

$$\lim_{N\to\infty} \beta_{N,r} = u_r \qquad \text{for every fixed } r. \tag{12.6}$$

This limit is unconditional: it is a finite algebraic calculation for the actual numbers, even when some weights are negative.

Degree elevation gives one further exact identity:

$$w_{N,k} = \frac{N+1-k}{N+1} w_{N+1,k} + \frac{k+1}{N+1} w_{N+1,k+1}. \tag{12.7}$$

It follows by applying ℓ to the identical relation for the Bernstein polynomials. Thus positivity at a higher level implies positivity at every lower level.

12.2 An exact positivity criterion and its fixed seed

Theorem 12.1 (Canonical grid criterion)**.** *For the actual moments in (12.1), the following statements are equivalent:*

(i) RH *holds.*

(ii) $w_{N,k} \geq 0$ *for every* $N \geq 0$ *and* $0 \leq k \leq N$.

(iii) There is a sequence of integers $N_j \to \infty$ *such that* $w_{N_j,k} \geq 0$ *for every* $0 \leq k \leq N_j$.

Under these conditions the weights construct a calibrated Hilbert seed with operator norm at most κ.

Proof. Under RH, Section 7 supplies $\lambda_j = \gamma_j^{-2} > 0$ with $\sum_j \lambda_j = \kappa$ and $m_r = \sum_j \lambda_j^{r+1}$. Hence $\nu = \sum_j (\lambda_j/\kappa)\delta_{\lambda_j/\kappa}$ is a probability measure on $[0,1]$ with moments u_r. Equation (12.2) becomes $w_{N,k} = \int b_{N,k}(t)\,\mathrm{d}\nu(t) \geq 0$, proving (i)⇒(ii). The implication (ii)⇒(iii) is immediate. Conversely, (12.7) propagates every positive level N_j downward, so (iii) implies (ii).

Assume (ii). On $\mathbb{C}^{N+1}$, for $N \geq 1$, define

$$K_N = \operatorname{diag}\left(\kappa\frac{k}{N}\right)_{k=0}^{N}, \qquad (v_N)_k = \sqrt{\kappa w_{N,k}}. \tag{12.8}$$

Then $0 \leq K_N \leq \kappa I$, $\|v_N\|^2 = \kappa$, and

$$\langle v_N, K_N^r v_N\rangle = \kappa^{r+1}\beta_{N,r} \longrightarrow m_r. \tag{12.9}$$

For each polynomial p, the quadratic forms of I, K_N, and $\kappa^2 I - K_N^2$ on $p(K_N)v_N$ are nonnegative. They involve only finitely many moments. Passing to the limit in each form gives (8.9)–(8.11) for the actual m_r, with the single bound $M = \kappa$. Theorem 8.1 reconstructs a bounded positive operator and a vector having all those moments. Its resolvent and the actual L have identical Taylor coefficients near zero, giving HSeed_Θ. Theorem 6.1 then proves RH. □

The fixed realization can also be displayed explicitly. Under (ii),

$$\nu_N = \sum_{k=0}^{N} w_{N,k}\delta_{k/N} \tag{12.10}$$

are probability measures on $[0,1]$. Compactness gives weakly convergent subsequences; (12.6) gives every limit the moments u_r. Polynomial density makes that limit unique, so the whole sequence converges to one probability measure ν. Consequently

$$\mathcal{H} = L^2([0,1],\nu), \qquad (Kf)(t) = \kappa t f(t), \qquad v(t) = \sqrt{\kappa} \tag{12.11}$$

is a fixed seed. Its scalar resolvent is $F(z) = \kappa \int (1+\kappa z t)^{-1}\,\mathrm{d}\nu(t)$, holomorphic on $D_\kappa = \mathbb{C} \setminus (-\infty, -1/\kappa]$. Taylor calibration and the analytic transfer identify F with the actual L throughout that domain.

12.3 A quantitative resolvent estimate

Under the positivity condition, the moments of ν imply $w_{N,k} = \int b_{N,k}(t)\,\mathrm{d}\nu(t)$. Thus the finite scalar resolvent

$$F_N(z) = \langle v_N, (I + zK_N)^{-1} v_N\rangle = \kappa \sum_{k=0}^{N} \frac{w_{N,k}}{1+\kappa z k/N} \tag{12.12}$$

is the integral of a binomial average of the resolvent kernel.

Proposition 12.2 (Explicit compact-domain error)**.** *Assume the equivalent conditions of Theorem 12.1. For a nonempty compact $C \subset D_\kappa$, set $R = \max_{z\in C}|z|$ and $\delta = \min_{z\in C,\,0\le t\le 1}|1+\kappa z t| > 0$. Then*

$$\sup_{z\in C}|F_N(z) - L(z)| \le \frac{\kappa^3 R^2 (u_1 - u_2)}{N\delta^3} \le \frac{\kappa^3 R^2}{4N\delta^3}. \tag{12.13}$$

For real $a \ge 0$, one also has $F_N(a) \ge L(a)$.

Proof. For fixed $t \in [0,1]$, let Y have distribution $\Pr(Y = k/N) = b_{N,k}(t)$. Then $\mathbf{E}(Y - t) = 0$ and $\mathbf{E}(Y-t)^2 = t(1-t)/N$. The exact scalar identity

$$\frac{1}{1+xy} - \frac{1}{1+xt} + \frac{x(y-t)}{(1+xt)^2} = \frac{x^2(y-t)^2}{(1+xt)^2(1+xy)} \tag{12.14}$$

with $x = \kappa z$ cancels its linear term after averaging in Y. Integrate against ν and multiply by κ. Each denominator factor has modulus at least δ, while $\int t(1-t)\,\mathrm{d}\nu = u_1 - u_2 \le 1/4$. This proves (12.13). For real $a \ge 0$ the right side of (12.14) is nonnegative, proving the last assertion. □

These finite models have vector normalization $\|v_N\|^2 = \kappa$, but $\operatorname{Tr} K_N = \kappa(N+1)/2$. Their calibrated moments are the weighted quantities (12.9), not $\operatorname{Tr}(K_N^{r+1})$. They therefore use the general Hilbert-seed criterion, without an identification with the trace-normalized determinant models. The possible grid atom at zero causes no limiting constant term: $w_{N,0} = \int (1-t)^N\,\mathrm{d}\nu \to \nu(\{0\}) = 0$. Here the last equality follows from calibration and the actual asymptotic $L(a) \to 0$ in (10.20), since $\lim_{a\to\infty} F(a) = \kappa\nu(\{0\})$.

Example 12.3 (A finite level need not extend)**.** For the test list $u_0 = 1$, $u_1 = 1/2$, $u_2 = 0$, level two has weights $(w_{2,0}, w_{2,1}, w_{2,2}) = (0,1,0)$. At level three, however, $w_{3,0} = -1/2 - u_3$ and $w_{3,3} = u_3$, so no choice of u_3 makes both nonnegative. The level-two grid has ordinary second moment $1/4$, not $u_2 = 0$, consistently with (12.5). The example concerns a test sequence, not the actual ξ coefficients.

The remaining input. The finite family, its moment convergence, and its conditional analytic error estimate are explicit. For the actual moments, the statement $w_{N,k} \geq 0$ at every level remains open in this construction. The downward identity (12.7) supplies no upward induction from a finite initial level. A finite raw-data checker would still need an independently proved mechanism establishing the entire positivity assertion from accepted finite evidence.

13 Exact finite moment extension and the remaining induction

There is a second exact description of the upward obligation, using the normalized actual logarithmic moments

$$u_r = \frac{m_r}{\kappa^{r+1}}, \qquad u_0 = 1. \tag{13.1}$$

The desired measure on $[0, \kappa]$ corresponds to a probability measure on $[0, 1]$ with moments u_r. We first describe a finite prefix without presupposing a representing measure. This is the classical truncated Hausdorff problem; the following derivation also records its singular cases [27, Section 4].

13.1 The exact finite positive-semidefinite conditions

For indices for which the entries are available, define

$$\begin{aligned} \mathsf{H}_d &= (u_{i+j})_{i,j=0}^{d}, & \mathsf{G}_d &= (u_{i+j+1} - u_{i+j+2})_{i,j=0}^{d}, \\ \mathsf{U}_d &= (u_{i+j+1})_{i,j=0}^{d}, & \mathsf{V}_d &= (u_{i+j} - u_{i+j+1})_{i,j=0}^{d}. \end{aligned} \tag{13.2}$$

Matrices with index -1 are empty. Positive semidefiniteness of an empty matrix is an empty condition.

Proposition 13.1 (Finite Hausdorff criterion)**.** *A real prefix $u_0, \ldots, u_N$, with $u_0 = 1$, has a representing probability measure on $[0, 1]$ if and only if*

$$\begin{cases} \mathsf{H}_d \succeq 0, & \mathsf{G}_{d-1} \succeq 0, & N = 2d, \\ \mathsf{U}_d \succeq 0, & \mathsf{V}_d \succeq 0, & N = 2d + 1. \end{cases} \tag{13.3}$$

Proof. Necessity follows by integrating $|p|^2$, $x(1-x)|p|^2$, $x|p|^2$, and $(1-x)|p|^2$ with the corresponding degree bounds. For sufficiency let $\mathcal{L}(x^j) = u_j$ on polynomials of degree at most N. We give the polynomial positivity argument explicitly.

If $p \geq 0$ on $[0, 1]$, then $R(t) = (1+t^2)^N p(t^2/(1+t^2))$ is an even nonnegative real polynomial of degree at most $2N$. Factoring its real roots with even multiplicity and pairing its nonreal roots gives $R = A^2 + B^2$, with $\deg A, \deg B \leq N$. Average this representation at t and $-t$, and separate even and odd parts. The result has the form $R(t) = \sum_j E_j(t^2)^2 + t^2 \sum_j O_j(t^2)^2$. Substitute $t^2 = x/(1-x)$ and multiply by $(1-x)^N$. For $N = 2d$ this gives $p = \sum a_j^2 + x(1-x)\sum b_j^2$, with $\deg a_j \leq d$, $\deg b_j \leq d-1$. For $N = 2d+1$ it gives $p = (1-x)\sum a_j^2 + x\sum b_j^2$, with both degrees at most d. These are polynomial identities, so the endpoint $x = 1$ is included. The conditions (13.3) imply $\mathcal{L}(p) \geq 0$ in either case.

The convex hull of $(x, x^2, \ldots, x^N)$, $0 \leq x \leq 1$, is compact. If $(u_1, \ldots, u_N)$ lay outside it, a separating affine functional would supply a polynomial $p \geq 0$ on $[0, 1]$ with $\mathcal{L}(p) < 0$. Hence the moment vector belongs to that convex hull, whose convex combinations give a representing probability measure. At most $N + 1$ atoms suffice by finite-dimensional Carathéodory. The case $N = 0$ is immediate. □

13.2 Two scalar bounds for the next prescribed moment

The block-matrix criterion needed here retains its range condition: for real symmetric $A \succeq 0$,

$$\begin{pmatrix} A & b \\ b^T & c \end{pmatrix} \succeq 0 \quad \Longleftrightarrow \quad b \in \mathrm{Ran}(A), \quad c \geq b^T A^\dagger b. \tag{13.4}$$

Here $A^\dagger$ is the Moore–Penrose inverse. Diagonalizing A shows that positivity forces the components of b in its nullspace to vanish; completing squares on the positive eigenspaces proves the remaining inequality and its converse. This also proves the singular case [27, Lemma 2.3]. An empty quadratic form contributes zero.

Suppose the prefix through u_{n-1} has a representing measure. For an even next index $n = 2d$, $d \geq 1$, set

$$\begin{aligned} A &= \mathsf{H}_{d-1}, & h &= (u_d, \ldots, u_{2d-1})^T, \\ C &= \mathsf{G}_{d-2}, & g &= (u_d - u_{d+1}, \ldots, u_{2d-2} - u_{2d-1})^T. \end{aligned} \tag{13.5}$$

For $d = 1$, C, g are empty. The exact allowable interval is

$$\underline{u}_{2d} = h^T A^\dagger h \;\leq\; u_{2d} \;\leq\; u_{2d-1} - g^T C^\dagger g = \overline{u}_{2d}. \tag{13.6}$$

For an odd next index $n = 2d + 1$, $d \geq 0$, set

$$\begin{aligned} A &= \mathsf{U}_{d-1}, & p &= (u_{d+1}, \ldots, u_{2d})^T, \\ C &= \mathsf{V}_{d-1}, & q &= (u_d - u_{d+1}, \ldots, u_{2d-1} - u_{2d})^T. \end{aligned} \tag{13.7}$$

For $d = 0$ both blocks and vectors are empty. The interval is

$$\underline{u}_{2d+1} = p^T A^\dagger p \;\leq\; u_{2d+1} \;\leq\; u_{2d} - q^T C^\dagger q = \overline{u}_{2d+1}. \tag{13.8}$$

Indeed, block-decompose the two matrices in (13.3) and apply (13.4). The old prefix's representing measure supplies at least one next moment. It therefore guarantees the requisite range conditions and that the interval is nonempty. Without that prefix hypothesis, positivity of the old blocks and both range conditions must be checked explicitly; the scalar endpoints alone do not suffice.

The first cases are

$$\begin{aligned} 0 \leq u_1 &\leq 1, \\ u_1^2 &\leq u_2 \leq u_1, \\ \frac{u_2^2}{u_1} &\leq u_3 \leq u_2 - \frac{(u_1 - u_2)^2}{1 - u_1} && (0 < u_1 < 1), \\ \frac{u_2^3 - 2u_1u_2u_3 + u_3^2}{u_2 - u_1^2} &\leq u_4 \leq u_3 - \frac{(u_2 - u_3)^2}{u_1 - u_2} && (u_1^2 < u_2 < u_1). \end{aligned} \tag{13.9}$$

Boundary cases use (13.4), without division by zero.

13.3 The precise induction obligation

Beginning with $u_0 = 1$, proving membership in the applicable interval for every *actual* u_n proves finite-prefix representability by induction. Compactness of probability measures on $[0, 1]$ then gives one measure representing all moments; polynomial approximation makes it unique. With $Kf(x) = \kappa x f(x)$ and $v(x) = \sqrt{\kappa}$, its resolvent has the actual Taylor coefficients $(-1)^r m_r$, hence agrees with L near zero. The existing analytic seed transfer then implies RH.

This describes the missing induction inequality rather than proving it. The extension theorem allows some next moment to be chosen inside a nonempty interval. The hyperbolic formulas already prescribe the actual next moment, whose membership still needs an independent proof. Ordinary positivity of the original source density does not establish these Schur inequalities for its nonlinear logarithmic-moment transform. The finite-witness project additionally needs a finite certificate whose soundness establishes the entire family of actual extension inequalities.

14 Actual cancellation budgets and the current branch

The previous sections describe positive realizations of the actual logarithmic moments. The current `RH_HILBERT_LIMIT` branch also measures the failure of positivity before it is known to vanish. This produces a precise semantic witness with just two natural-number fields. The distinction between its finite data and its infinite admissibility condition is essential.

14.1 Exact carriers, variation and refinement

Keep $u_n = m_n/\kappa^{n+1}$ and the actual weights $w_{n,j}$ of (12.1)–(12.2). The implementation uses the entire even descent `AWPIRH.theta`, the classical normalized completion `AWPIRH.xi`, and the complex derivatives of its logarithmic derivative. The complex moments are proved real, and their real coordinates define the moments used here. In particular, the definition of u_n is tied to the actual ξ, not an arbitrary model sequence. The source completion includes the factor $1/2$ in (1.2).

Define the row variation and the local cancellation defect by

$$V_n = \sum_{j=0}^{n} |w_{n,j}|, \tag{14.1}$$

$$\begin{aligned} d_{n,j} &= (n+1-j)|w_{n+1,j}| + (j+1)|w_{n+1,j+1}| \\ &\quad - |(n+1-j)w_{n+1,j} + (j+1)w_{n+1,j+1}|\,, \end{aligned} \tag{14.2}$$

$$C_n = \sum_{j=0}^{n} d_{n,j}. \tag{14.3}$$

The symbol V_n in this section denotes a real scalar variation; it is distinct from the Hilbert vector V_a in (1.1).

Proposition 14.1 (Exact cancellation identity)**.** *For the actual normalized rows,*

$$V_0 = 1, \qquad V_n \geq 1, \qquad C_n = (n+1)(V_{n+1} - V_n) \geq 0. \tag{14.4}$$

Thus V_n is nondecreasing. Moreover $V_n = 1$ if and only if every weight in row n is nonnegative.

Proof. The row mass is 1 by (12.3), so the triangle inequality gives $V_n \geq 1$; row zero is the singleton $u_0 = 1$. Each $d_{n,j}$ is nonnegative by the real triangle inequality. Multiplying degree elevation (12.7) by $n+1$ identifies the last absolute value in (14.2) with $(n+1)|w_{n,j}|$. In the sum of the first two terms, each interior entry $|w_{n+1,j}|$ has total coefficient $(n+1-j)+j = n+1$; the two endpoints have that same coefficient. Summing therefore gives (14.4). Finally $\sum_j |w_{n,j}| - \sum_j w_{n,j} = 2\sum_j \max(-w_{n,j}, 0)$. This vanishes exactly when the finite row has no negative entry. □

These are the objects `actualVariation` and `rowCancellation actualNormalizedMoment` in the source. The definitions and identity are retained in RH_HilbertLimit/Cancellation.lean and its imported Bernstein calculus. No positive-measure assumption is used to define them.

14.2 The ordinary root limit is already an exact endpoint

The branch constructs multiplicity-bearing actual zero scales a_i and canonical nodes $q_i = (\kappa a_i)^{-1}$. Its exact power-sum calibration is

$$u_n = \sum_i q_i^{n+1}, \qquad \sum_i q_i = 1, \tag{14.5}$$

with the convergence and reality statements supplied by the imported actual-function construction. These nodes need not be assumed real or positive. The relevant spectral radius is

$$R = \sup\bigl(\{1\} \cup \{\, |q_i| + |1 - q_i| : i \,\}\bigr). \tag{14.6}$$

The constructed declarations in RH_HilbertLimit/AnalyticClosure.lean give the ordinary limit and its RH specializations:

$$\lim_{n\to\infty} V_n^{1/n} = R \geq 1, \tag{14.7}$$

$$\mathrm{RH} \iff R = 1 \iff (\forall n,\ V_n = 1), \tag{14.8}$$

$$\mathrm{RH} \iff \forall \varepsilon > 0\ \exists A > 0\ \forall n, \quad V_n \leq A e^{\varepsilon n}. \tag{14.9}$$

Here n tends through positive integers. The Lean function `rootScale` uses exponent $(n : \mathbb{R})^{-1}$; its value at index zero does not alter the limit. Equation (14.7) is an ordinary limit, not just a limsup. Its existence alone is not RH: one still has to force its value to be 1.

The exact names are canonical_actual_root_tendsto_constructed, canonical_actual_root_one_iff_nativeRH, canonical_actual_subexponential_iff_nativeRH, and canonical_actual_all_variation_one_iff_nativeRH. Their constructed versions have no extra calibration or dominant-extraction hypothesis. They instantiate the generic spectral results with the actual calibration and extraction developed in this branch. This paragraph records that source endpoint and its historical audit boundary; it does not promote the entire manuscript to a newly compiled formal proof.

14.3 Finite data with a universal admissibility condition

The structure `CancellationBudget` contains exactly `cutoff : Nat` and `coefficient : Nat`. Write it as $W = (N, k) \in \mathbb{N}^2$, using lower-case k to distinguish the coefficient from the Hilbert operator K. Its separate admissibility predicate is

$$\mathrm{Adm}(N, k) \quad :\iff \quad \forall n \in \mathbb{N}, \quad N \leq n \implies C_n \leq k V_n. \tag{14.10}$$

The predicate is a proposition, not a Boolean procedure. Its all-order quantifier is not stored as an extra field in the finite pair.

Proposition 14.2 (Growth consequence of a cancellation budget)**.** *If* (14.10) *holds, then, for every* $n \geq N$*,*

$$V_{n+1} \leq \left(1 + \frac{k}{n+1}\right) V_n, \qquad V_n \leq V_N \left(\frac{n+1}{N+1}\right)^k. \tag{14.11}$$

Consequently V *satisfies the subexponential condition in* (14.9)*.*

Proof. The first inequality follows from (14.4) and division by the positive number $n+1$. Since k is a natural number, Bernoulli's inequality gives $1 + k/(n+1) \leq (1 + 1/(n+1))^k$. Iterating from N to $n-1$ telescopes the factors $((j+2)/(j+1))^k$ and proves the second inequality. A fixed polynomial is bounded by a constant times $e^{\varepsilon n}$ for every $\varepsilon > 0$; enlarging that positive constant covers the finitely many indices below N. □

Combining this calculation with the actual endpoint (14.9), and using RH to obtain $V_n = 1$, gives

$$\boxed{(\exists N, k \in \mathbb{N},\ \mathrm{Adm}(N,k)) \Longleftrightarrow \mathrm{RH}.} \tag{14.12}$$

The converse chooses $(N,k) = (0,0)$: constant variation gives $C_n = 0$ at every index. In Lean, the exact declaration is RH_HilbertLimit.exists_admissible_cancellationBudget_iff_nativeRH. The forward declarations include nativeRH_of_admissible_cancellationBudget and nativeRH_of_exists_admissible_cancellationBudget. The source is RH_HilbertLimit/ActualCancellationWitness.lean.

The retained v0.2.20 user-supplied native output explicitly lists only `propext`, `Classical.choice`, and `Quot.sound` for the semantic endpoint. The v0.2.23 user-supplied record reports aggregate reuse and audit PASS for 394 selected declarations. These historical records are not new runs performed for this paper, and local Windows anchor paths alone do not supply their receipt contents.

14.4 The zero budget and the missing upward theorem

For the explicit candidate $W_0 = (0,0)$, nonnegativity of cancellation gives

$$\begin{aligned} \mathrm{Adm}(W_0) &\Longleftrightarrow \forall n,\ C_n = 0 \\ &\Longleftrightarrow \forall n,\ V_n = 1 \Longleftrightarrow \mathrm{RH}. \end{aligned} \tag{14.13}$$

An equivalent induction formulation is

$$C_0 = 0 \quad \text{and} \quad \forall n,\ (C_n = 0 \Longrightarrow C_{n+1} = 0). \tag{14.14}$$

Induction proves universal vanishing from (14.14); the reverse implication is immediate. This logical equivalence does not prove its upward premise. In particular, the degree-elevation identity propagates positivity downward from a higher row; it is not an upward positivity theorem.

The reported successful v0.2.24 native checkpoint supports $0 < u_1 \le 1$, $V_1 = 1$, and $C_0 = 0$, with its explicit refinement scope limited to index zero. The first nontrivial extension seeks the actual second-moment inequalities

$$0 \le u_2, \qquad u_2 \le u_1, \qquad 2u_1 \le 1 + u_2, \tag{14.15}$$

which make the degree-two row nonnegative and hence give $C_1 = 0$ once $V_1 = 1$ is known. Their current Lean sources still await a successful native compilation of the repaired module. Choosing the pair $(0,0)$ does not discharge (14.13).

14.5 The first-five source construction and its finite scope

The current proposed endpoint is the precise finite statement

$$\forall n \in \mathbb{N}, \qquad n < 5 \Longrightarrow C_n = 0. \tag{14.16}$$

It is named zeroBudgetCandidate_first_five_cancellations in RH_HilbertLimit/ZeroBudgetFiveRows.lean. The construction seeks a cubic theta-kernel envelope, then the strip estimate $\|\mathrm{completedRiemannZeta}_0(s)\| \le 9/190$ for $0 \le \Re s \le 1$. Together with $\|s(s-1)\| \le 305/16$ for $|\Im s| \le 17/4$, the product bound is $549/608 < 1$, excluding actual ξ zeros in that finite-height region. The resulting canonical-node estimates are

$$4|\Im q_i| \le \Re q_i, \qquad 4|\Im q_i| \le 1 - \Re q_i. \tag{14.17}$$

Exact rational cone multiplication through six factors then targets nonnegative real parts of $q_i^{j+1}(1-q_i)^{d-j}$ for $d \le 5$. The calibrated spectral sums transfer these inequalities to the actual Bernstein weights. This would give $V_d = 1$ through degree five and (14.16). This paragraph describes the proof-source construction; it is not a successful native first-five audit.

The latest supplied failed-run tail in this branch is the v0.2.27 run. It reports failures of `ZeroBudgetFiveSector` and `ZeroBudgetSecondMoment`, and blocks `ZeroBudgetFiveRows`. The v0.2.28 `SECOND_MOMENT_FIX` package removes two redundant `ring` calls after already-completed square-identity proofs. The actual `FiveSector` error was absent from the supplied excerpt; that module is unchanged and its failure is unresolved. No native run of the partial repair is included in this revision.

The package contains 106 Lean source files, 99 ordinary build modules and one selected final audit, with 463 selected audit targets. These are inventory counts, not a claim that all current sources passed. A first-five PASS, if later obtained, would establish only indices $0, 1, 2, 3, 4$. Indices 5–9 are not included, and universal propagation remains open.

14.6 The finite-checker gap after the semantic equivalence

Equation (14.12) is stronger evidence about the exact mathematical admission condition, but is not the finite-checker endpoint (15.1). A route through cancellation still needs one explicit terminating checker on finite raw data, with proved implications

$$\mathsf{check}(W) = \mathsf{true} \Longrightarrow \mathrm{Adm}(N(W), k(W)), \tag{14.18}$$

$$\mathrm{RH} \Longrightarrow \exists W,\ \mathsf{check}(W) = \mathsf{true}. \tag{14.19}$$

The first line must derive the universal bound, not accept it as an unverified proof field or an oracle. The second line must refer to the same independently specified checker. Neither is obtained merely by observing that the semantic budget has two natural-number fields. No such accepted finite witness is supplied here.

Current mathematical boundary. The semantic admissible-budget equivalence has historical user-reported native evidence. The first zero-budget refinement has a reported successful checkpoint. The first-five extension is a source candidate under repair. Universal zero-budget admissibility, terminating-checker soundness and completeness, and unconditional RH remain open.

15 Why a finite matrix is not the finite witness

15.1 An obstruction to exact finite-dimensional calibration

Proposition 15.1 (Finite visible spectrum is impossible)**.** *There is no exactly calibrated seed for the actual Θ whose scalar spectral measure has finite support. In particular, no finite-dimensional or finite-rank operator can supply an exact seed.*

Proof. A finitely supported spectral measure makes $F(z) = \sum_{j=1}^{q} c_j/(1+\lambda_j z)$ rational, with a possible constant term when $\lambda_j = 0$. Write $F = p/q_0$ for polynomials, where q_0 is not identically zero. Exact interval calibration implies that the entire function

$$q_0(z)\Theta'(z) - p(z)\Theta(z)$$

vanishes on an interval and therefore vanishes on all of $\mathbb{C}$. At any zero z_0 of Θ with $q_0(z_0) \neq 0$, the local ODE $\Theta' = (p/q_0)\Theta$ contradicts the finite-order zero argument of Theorem 6.1. Thus

every zero of Θ would have to be among the finitely many roots of q_0. But ξ has infinitely many distinct zeros, and the two-to-one square map can identify only opposite pairs. Hence Θ has infinitely many distinct zeros, a contradiction.

If K is finite rank, its nonzero spectral values form a finite set; its kernel can contribute only the additional spectral point zero. Its scalar spectral measure is therefore finitely supported, so the same argument applies. □

This obstruction does not prohibit a finite description of an infinite-dimensional operator or a finite certificate for an infinite analytic argument. It rules out the narrower proposal that a finite positive matrix by itself can reproduce the full actual logarithmic derivative exactly on an interval.

15.2 The two missing checker theorems

The project's requested finite endpoint is

$$(\exists W : \mathsf{FiniteWitness},\ \mathsf{check}(W) = \mathsf{true}) \quad \Longleftrightarrow \quad \mathrm{RH}. \tag{15.1}$$

The witness must consist of finite raw data and the checker must terminate. The analytic seed defined in this paper is not such a witness type: it quantifies over a Hilbert space, an operator, a vector, and an exact interval identity.

A route through Theorem 1.3 would require the following separate statements for one specified checker:

$$\mathsf{check}(W) = \mathsf{true} \implies \mathsf{HSeed}_{\Theta}, \qquad \text{for every finite } W; \tag{15.2}$$

$$\mathrm{RH} \implies \exists W,\ \mathsf{check}(W) = \mathsf{true}. \tag{15.3}$$

Equation (15.2) is actual-function soundness. It must derive calibration, not receive an unverified calibration proof as a field of the purported raw witness. Equation (15.3) is finite-language completeness. Constructing a countable spectrum from all zeros under RH proves an analytic existence statement; it does not extract an accepted finite code in an independently fixed language.

Finite moment matrices, finite Gram matrices and finite truncations are legitimate components of a possible certificate design. To yield (15.2), they need a theorem extending the verified finite information to the uniform analytic assertion. The all-order moment criterion in Corollary 8.2 specifies one precise target for such an extension. No such finite reduction is established by the present note.

15.3 Finite jets cannot replace a functional identity

There is a direct mathematical reason for the distinction. Fix a real anchor $a_* > 0$ and an integer $N \geq 0$. The functions

$$L(z) \quad \text{and} \quad L(z) + \varepsilon(z - a_*)^{N+1}$$

have identical derivatives through order N at a_* for every ε, but are different analytic functions when $\varepsilon \neq 0$. Finitely many samples can likewise be preserved by adding a nonzero polynomial vanishing at all sample points. The identity theorem requires a set with an accumulation point in the analytic domain, or equivalent exact all-order germ information. A finite initial jet is neither of these.

16 Repository state and the evidence boundary

The earlier analytic discussion was prepared against `randogelist/Riemann-Hypothesis-Final` at revision `362d9f8aeaaf9479cc27ceb6b63bdb7456267990`. Its slit-plane ODE admission, actual zero geometry and positive-product assessment remain relevant [7, 8, 9]. The source admission uses $\Theta' + A\Theta = 0$, while the paper uses $\Theta' = F\Theta$, so their identification is $A = -F$.

This revision preserves that derivation and publishes the current Hilbert-limit source snapshot on branch `CERTIFICATE`, based on repository commit ba58f416746517b865c06ebf64e75b283fbfe437. The revised PDF is in the repository root and its editable source is under docs/papers/. The branch snapshot, native runner, audit manifests and historical evidence are under research/certificate/. That snapshot is an explicitly separate native-Windows project; publication does not silently replace the repository's older default build plan.

The manuscript's Hilbert seed definitions are mathematical formulations, not a claim of field-for-field identity with a legacy `HasSeed`. The cancellation-budget section instead names the exact declarations and carriers present in the current branch snapshot.

Layer	Statement	Evidence and limit
Hilbert criterion	Fixed calibrated resolvent iff RH	Retained on-paper derivation; no manuscript-wide Lean audit
Hyperbolic/Jensen routes	Model identities, actual coefficient limits, extension conditions	On-paper results; all-order positivity or calibration obligations remain
Actual root limit	Ordinary limit $V_n^{1/n} \to R$; $R = 1$ iff RH	Constructed Lean declarations; historical user-reported native evidence
Semantic budget	$\exists N, k : \mathrm{Adm}(N, k)$ iff RH	v0.2.20 explicit endpoint axiom output; v0.2.23 aggregate reuse PASS
Zero-budget base	$0 < u_1 \le 1$, $V_1 = 1$, $C_0 = 0$	User-reported v0.2.24 PASS; index zero only
First-five extension	$C_n = 0$ for $n < 5$	Current source candidate; v0.2.27 build failed
Latest repair	Two redundant tactics removed in SecondMoment	v0.2.28 partial source repair; not natively run; FiveSector unresolved
Universal admissibility	$C_n = 0$ at every index for $(0, 0)$	Open; finite prefix does not prove the upward rule
Finite checker	Terminating acceptance, actual soundness, finite completeness	Open; no accepted actual-RH witness

Historical native evidence is retained as user-supplied log transcriptions and metadata, with its package and declaration scope. In particular, evidence/native_pass_v0221/USER_NATIVE_V0220_PASS_TAIL.txt contains the explicit endpoint axiom lists; evidence/zero_budget_v0224

/USER_NATIVE_V0223_PASS_EXCERPT.txt retains the later aggregate successful run; and evidence/zero_budget_second_v0225/USER_NATIVE_V0224_PASS_EXCERPT.txt records the first refinement. The partial repair records the latest failure under evidence/no_goals_fix_v0228/. These records do not assert access to the user's local Windows receipt files or to independently recovered original log bytes.

The execution target remains `LOCAL_COMPILER`, with native Lean 4.33.0, compatible cached Mathlib and the existing `LEAN_ANCHORINGS` validation rules. Source, dependency, toolchain, artifact and same-target prior PASS fingerprints govern reuse. This PDF update invokes no Lean compiler, Lake, native proof runner or GitHub Actions. The name `CERTIFICATE` labels a Git branch; it is not proof certification.

17 Conclusion

The Hilbert-vector anchor has an explicit on-paper analytic formulation. Its central scalar identity is

$$\frac{\mathrm{d}}{\mathrm{d}a}\left(a\frac{\Theta'(a)}{\Theta(a)}\right) = \|(I+aK)^{-1}v\|^2,$$

where K and v are fixed, K is bounded and nonnegative self-adjoint, and the identity is exact on a positive interval. The proof proceeds through a holomorphic resolvent, an explicitly removed integration constant, and a division-free analytic ODE. The zero-order contradiction then yields the critical-line statement.

Conversely RH supplies a diagonal seed through the genus-zero product of the entire square-descended function. This gives the analytic equivalence (1.6), the Gram identity, and the normalization (7.7). It does not construct the seed without RH. The unresolved constructive task is to derive one of these exact analytic inputs from independently verifiable data; the further finite-checker equivalence requires both (15.2) and (15.3).

Section 10 gives a precise role for the Hyperbolic Zeta identities. The Bose weights construct positive bounded model resolvents and exact odd-zeta formulas; their asymptotics exclude a direct identification with the actual L. A separate positive hyperbolic-tangent integral determines actual ξ jets, and the Newton recurrence converts those jets to the required logarithmic moments. Squaring the project's signed pencils supplies positive determinant models. Theorem 10.5 proves that calibration of all their coefficients, or all their trace moments, would construct the fixed seed and imply RH. That all-order limit theorem, and a completeness theorem for this particular family, remain open.

Sections 11–13 make the next construction more explicit. The canonical polynomials P_n converge to the actual $\Theta/\Theta(0)$ without RH; the remaining condition is their hyperbolicity, or the equivalent actual Hermite-matrix positivity, at unbounded degrees. The Appell derivative identity reduces a proposed upward step to a precise interval for the next actual coefficient. A second canonical route supplies grid resolvents from the actual logarithmic moments, with exact moment limits and an explicit error bound conditional on positive weights. The Schur formulas isolate the corresponding next-moment intervals. None of these constructions proves that the actual prescribed coefficient belongs to its required interval at every order.

The available base does not supply an upward theorem. Section 14 records the semantic equivalence between an admissible two-natural-number cancellation budget and RH, with historical native evidence separate from the on-paper derivations. For $(0,0)$ the reported base is $C_0 = 0$; universal propagation remains open. The first-five sources are under native repair. This revision supplies no terminating checker, accepted finite witness, new native build or unconditional RH proof.

A Real Hilbert spaces and the holomorphic convention

The recovered anchor may be expressed on a real Hilbert space. That does not obstruct the complex-analytic proof. If H is real, form $H_{\mathbb{C}} = H \oplus iH$ with

$$\langle x + iy, u + iw\rangle_{\mathbb{C}} = (x,u)_H + (y,w)_H \\ + i\big((x,w)_H - (y,u)_H\big).$$

This inner product is conjugate-linear in the first slot and linear in the second. A real bounded self-adjoint nonnegative K extends by $K_{\mathbb{C}}(x+iy) = Kx + iKy$. Its norm is unchanged, because

$$\|K_{\mathbb{C}}(x+iy)\|^2 = \|Kx\|^2 + \|Ky\|^2 \le \|K\|^2(\|x\|^2 + \|y\|^2),$$

and restriction to real vectors gives the reverse norm inequality. Self-adjointness follows by expanding the inner product. Positivity follows from

$$\langle x + iy, K_{\mathbb{C}}(x+iy)\rangle = (x,Kx)_H + (y,Ky)_H \ge 0,$$

since the imaginary cross terms cancel by symmetry. For real $a \ge 0$, the inverse extends the real inverse, so every real scalar calibration and norm identity is preserved. Thus the complex theorem applies to the real Hilbert version as well.

If an alternative convention makes the inner product linear in the first slot, the scalar holomorphic expression should instead be $\langle R_z v, v\rangle$. With the convention fixed in this paper, it is $\langle v, R_z v\rangle$. Mixing these conventions would turn a holomorphic function into an anti-holomorphic one and invalidate the identity-theorem step.

B A compact proof dependency ledger

The forward implication has the following dependencies, each proved or cited at its point of use:

- **F1.** The actual Θ is entire and $\Theta(0) > 0$.
- **F2.** Fixed bounded positive K gives a holomorphic resolvent on D_M, a connected domain containing zero and the positive ray.
- **F3.** Exact interval calibration gives $\Theta' = F\Theta$ by the identity theorem. For differential calibration, first remove C using $Q(0) = -C\Theta(0)$.
- **F4.** A nonzero entire solution of that ODE has no zero where its coefficient is holomorphic, by the finite-order contradiction.
- **F5.** Negative-real-axis confinement of Θ zeros is equivalent to the critical-line condition for actual ξ zeros.

Only the converse uses RH:

- **R1.** RH places the zeros of Θ at $-\gamma_n^2$.
- **R2.** The order bound and Hadamard factorization give $\Theta/\Theta(0) = \prod_n (1 + z/\gamma_n^2)$ and summability.
- **R3.** Uniformly convergent logarithmic differentiation gives $L(a) = \sum_n (a + \gamma_n^2)^{-1}$.
- **R4.** The explicit diagonal K and square-summable v realize this scalar function and hence both seed conditions.

The all-order moment criterion reconstructs the forward input through Theorem 8.1. It does not remove an infinite quantifier by itself. Neither direction in this ledger uses a finite accepted witness, and neither is reported as a Lean-verified endpoint.

C Spectral restrictions on the original squared pencils

The normalized squares in (10.43) have additional structure beyond positivity. Let $\beta_{n,j}$ be the nonzero eigenvalues of B_n, and arrange

$$p_{n,j} = \frac{\beta_{n,j}^2}{\operatorname{Tr}(B_n^2)} \tag{C.1}$$

in decreasing order, padding each list with zeros. Then $p_{n,j} \geq 0$, $\sum_j p_{n,j} = 1$, and

$$Q_n(z) = \prod_j (1 + \kappa p_{n,j} z), \qquad \frac{\operatorname{Tr}(K_n^{r+1})}{\kappa^{r+1}} = \sum_j p_{n,j}^{r+1}. \tag{C.2}$$

Thus the squared-pencil problem asks for a particular sequence of finite probability partitions with the prescribed actual power sums.

Proposition C.1 (Subsequential determinant limits)**.** *Every sequence in* (C.1) *has a subsequence and a decreasing sequence* $(p_j)_{j\geq 1}$ *with* $p_j \geq 0$ *and* $\sum_j p_j \leq 1$ *such that, locally uniformly on* $\mathbb{C}$*,*

$$Q_n(z) \longrightarrow Q_\infty(z) := e^{\kappa d z} \prod_{j\geq 1} (1 + \kappa p_j z), \qquad d = 1 - \sum_{j\geq 1} p_j \geq 0. \tag{C.3}$$

Here n *ranges over the extracted subsequence.*

Proof. Since $p_{n,j} \leq 1/j$, diagonal subsequence extraction gives $p_{n,j} \to p_j$ for every j. Taking limits in each finite partial sum gives $\sum_j p_j \leq 1$. For every integer $r \geq 1$,

$$\sum_{j>J} p_{n,j}^{r+1} \leq p_{n,J+1}^{r} \sum_{j>J} p_{n,j} \leq (J+1)^{-r}. \tag{C.4}$$

The same bound holds for the limiting sequence. Therefore $\sum_j p_{n,j}^{r+1} \to \sum_j p_j^{r+1}$, while the first power sum of every finite partition remains 1.

The product in (C.3) converges locally uniformly because $\sum_j p_j < \infty$. In $|\kappa z| < 1$, its logarithm with value zero at the origin is

$$\log Q_\infty(z) = \kappa z + \sum_{q=2}^{\infty} \frac{(-1)^{q-1} (\kappa z)^q}{q} \sum_j p_j^q.$$

The series is absolutely convergent. Formula (C.4) shows that all finite power sums of K_n converge to the values in this logarithm. Newton's identities, as used in Theorem 10.5, now give coefficientwise convergence of Q_n to Q_∞. The bounds $0 \leq [z^q]Q_n \leq \kappa^q/q!$ pass to the limit. On each disk $|z| \leq R$, the common majorant $\sum_q (\kappa R)^q/q!$ turns coefficientwise convergence into uniform convergence. □

The term d has a precise measure interpretation. Along this subsequence, the probability measures $\sum_j p_{n,j}\delta_{p_{n,j}}$ converge weakly to $d\delta_0 + \sum_j p_j \delta_{p_j}$. Their masses are one and their positive-degree moments converge by (C.4); polynomial approximation on $[0,1]$ proves the weak convergence. The atom at zero records mass carried by entries which individually tend to zero.

Corollary C.2 (Necessary spectral behavior for actual calibration)**.** *If* $Q_n \to \Theta/\Theta(0)$ *locally uniformly, then every extracted limit above satisfies*

$$d = 0, \qquad \sum_j p_j = 1, \qquad \#\{j : p_j > 0\} = \infty. \tag{C.5}$$

In particular $\operatorname{rank} B_n \to \infty$*.*

Proof. The unconditional order bound used in Section 7 is $\operatorname{ord}\Theta \le 1/2$. Since $\Theta(a) > 0$ for $a \ge 0$, it gives $\limsup_{a\to\infty} a^{-1}\log(\Theta(a)/\Theta(0)) \le 0$. On the other hand, every factor in (C.3) is at least one on that ray, so $\log Q_\infty(a) \ge \kappa d a$. Hence $d = 0$. If only finitely many p_j were positive, Q_∞ would be a polynomial. This contradicts the infinitely many distinct zeros of the actual Θ. Finally, a subsequence of bounded ranks would have a further extracted partition limit with only finitely many nonzero entries, which has just been excluded. □

A quantitative exclusion of flat spectra

Set

$$\delta_n = \max_j p_{n,j}, \qquad \eta_n = \sum_j p_{n,j}^2 = \frac{\operatorname{Tr}(B_n^4)}{(\operatorname{Tr}(B_n^2))^2}.$$

The inequalities $\delta_n^2 \le \eta_n \le \delta_n$ follow from the largest summand and $\sum_j p_{n,j} = 1$. For $|z| \le R$ with $\kappa R\delta_n < 1$, choose the logarithm of each factor from the origin. The power series for $\log(1+w)$ gives $|\log(1+w) - w| \le |w|^2/[2(1-|w|)]$, and therefore

$$|\log Q_n(z) - \kappa z| \le \frac{(\kappa R)^2\eta_n}{2(1-\kappa R\delta_n)} =: E_n(R), \tag{C.6}$$

$$\sup_{|z|\le R} |Q_n(z) - e^{\kappa z}| \le e^{\kappa R}(e^{E_n(R)} - 1). \tag{C.7}$$

Consequently $\eta_n \to 0$, equivalently $\delta_n \to 0$, forces $Q_n \to e^{\kappa z}$ locally uniformly, and excludes actual calibration.

For example, let $r_n = \operatorname{rank} B_n \to \infty$. If all nonzero eigenvalues satisfy $\max_j |\beta_{n,j}|/\min_j |\beta_{n,j}| \le C$ for one fixed C, then $\delta_n \le C^2/r_n \to 0$. This parameter regime is excluded by (C.7). No assertion that a particular admissible Hyperbolic Zeta schedule satisfies this bounded-ratio hypothesis is being made.

Exact diagnostics from the pencil determinant

For one pencil (10.39), let $g(u) = \log(\Delta(x_0+u)/\Delta(x_0))$ be the local analytic logarithm with $g(0) = 0$. Since $\Delta(x_0+u)/\Delta(x_0) = \det(I + uB)$, differentiation yields

$$g^{(q)}(0) = (-1)^{q-1}(q-1)!\operatorname{Tr}(B^q), \qquad q \ge 1. \tag{C.8}$$

Thus for every integer $r \ge 0$,

$$\frac{\operatorname{Tr}(B^{2r+2})}{(\operatorname{Tr}(B^2))^{r+1}} = \frac{-g^{(2r+2)}(0)}{(2r+1)!\,(-g''(0))^{r+1}}. \tag{C.9}$$

The denominator is positive. These are exact formulas for the trace targets directly from Δ, without constructing $G^{-1/2}$ explicitly.

For the original admissible pencils B is invertible. Their real roots, counted with multiplicity, satisfy $x_j = x_0 - 1/\beta_j$. Consequently

$$p_j = \frac{|x_0 - x_j|^{-2}}{\sum_\ell |x_0 - x_\ell|^{-2}}. \tag{C.10}$$

A bounded ratio of these root distances implies the bounded absolute eigenvalue ratio above. The roots x_j belong to the zeta-value parameter X of the affine pencil. Their identification with data from the complex argument of the actual ξ has not been established. The restrictions and identities in this appendix give tests for a proposed schedule; the actual all-order trace limits in Section 10.7 remain an open obligation.

References

[1] NIST Digital Library of Mathematical Functions, *Section 25.4: Reflection Formulas*, especially Eqs. 25.4.3–25.4.4. https://dlmf.nist.gov/25.4. Accessed 27 October 2026.

[2] NIST Digital Library of Mathematical Functions, *Section 25.2: Definition and Expansions*, especially Eq. 25.2.3. https://dlmf.nist.gov/25.2. Accessed 27 October 2026.

[3] NIST Digital Library of Mathematical Functions, *Section 25.10: Zeros*, subsection (i). https://dlmf.nist.gov/25.10. Accessed 27 October 2026.

[4] Paul Garrett, *Riemann-Hadamard product for* $\zeta(s)$, 24 October 2021, pp. 1–2 (growth of the completed entire function). https://www-users.cse.umn.edu/~garrett/m/number_theory/riemann_hadamard_product.pdf.

[5] C. McMullen, *Advanced Complex Analysis*, Harvard Math 213a course notes, 22 January 2017. Theorems 3.7 and 3.11, printed pp. 43–44 (Hadamard factorization and zero summability). https://people.math.harvard.edu/~ctm/home/text/class/harvard/213a/10/html/home/course/course.pdf.

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[7] Riemann-Hypothesis-Final project, `SoloboloAdmission/NativeODE.lean`, source revision `362d9f8`. Exact source of `slitODEAdmission_iff_RH`; native audit not run in this note. Pinned repository source.

[8] Riemann-Hypothesis-Final project, `AngularMaster/ZeroGeometry.lean`, source revision `362d9f8`. Pinned repository source.

[9] Riemann-Hypothesis-Final project, *RH positive-product assessment*, 25 October 2026, source revision `362d9f8`. Pinned research assessment.

[10] Riemann-Hypothesis-Final project, `provenance/status.json`, `docs/BUILD.md`, and `docs/FRONTIER.md`, source revision `362d9f8`. Source evidence and open finite calibration obligations. Pinned status record.

[11] Prince Anand, *Spectral Theory and Diophantine Bounds for Hankel Determinants of Odd Zeta Values*, supplied project manuscript, printed pp. 7–9, especially Eq. (3.1) and Theorem 3.1. Tracked in `randogelist/zetahyperbolic-axiom`, revision `b56c8102d25b931a601992c93bf30b19cc7ee103`. Pinned manuscript repository. Only the identities identified in Section 10 are used here; other manuscript endpoints are outside this note's scope.

[12] zetahyperbolic-axiom project, `Theorem_3_1_Hyperbolic_Weight.lean`, revision `b56c810`, declarations `density` and `density_pos`. Source inspection; no fresh native audit. Pinned weight source.

[13] zetahyperbolic-axiom project, `Theorem_3_1_Hyperbolic_Moments.lean`, revision `b56c810`, declaration `integral_monomial_density`. Pinned moment source.

[14] zetahyperbolic-axiom project, `Theorem_3_1_Pole_Integral_Identity.lean`, revision `b56c810`, declaration `poleIntegralIdentity`. Its integer-parameter statement is extended on paper to real positive parameters in Proposition 10.2. Pinned pole-integral source.

[15] zetahyperbolic-axiom project, `LeanPaper/ContinuousDensityNormalization.lean`, revision `b56c810`, declarations `correctedDensity_odd` and `printedDensity_odd`. Pinned normalization correction.

[16] zetahyperbolic-axiom project, `LeanPaper/ContinuousGram.lean`, revision `b56c810`, declarations `baseMatrix_entry_exact`, `baseMatrix_posDef`, and `pencil_odd`. Pinned Gram source.

[17] zetahyperbolic-axiom project, `LeanPaper/PoleWeightSigns.lean`, revision `b56c810`, declarations `poleWeight_sign` and `gamma_has_both_signs`. Pinned sign source. See also `snditeration`, revision `35218ba6ee152774ff4e1fb778e530c1752acfd6`, `Zetahyperbolic/Section3/SpectralInertia.lean`, `actual_inertia_formula`.

[18] NIST Digital Library of Mathematical Functions, *Section 5.9: Integral Representations*, Eq. 5.9.15 (Binet's digamma formula). https://dlmf.nist.gov/5.9. Accessed 27 October 2026.

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[20] NIST Digital Library of Mathematical Functions, *Section 27.4: Euler Products and Dirichlet Series*, Eq. 27.4.12. https://dlmf.nist.gov/27.4. Accessed 27 October 2026.

[21] NIST Digital Library of Mathematical Functions, *Section 25.5: Integral Representations*, Eq. 25.5.3. https://dlmf.nist.gov/25.5. Accessed 27 October 2026.

[22] NIST Digital Library of Mathematical Functions, *Section 5.5: Functional Relations*, Eq. 5.5.5 (gamma duplication). https://dlmf.nist.gov/5.5. Accessed 27 October 2026.

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